

LIGO Detector Characterization in the Second and Third Parts of the Fourth Observing Run

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Abstract. LIGO detector characterization efforts enabled the confident detection of gravitational waves from hundreds of compact binary coalescences during the fourth observing run. Reliable production of high quality detector data and rapid noise mitigation efforts allow the extraction of the most in-depth knowledge of gravitational wave sources and their progenitors. In this paper we describe LIGO detector characterization activities during the second and third parts of O4—O4b and O4c. We summarize changes in detector configuration and performance at the LIGO Hanford and LIGO Livingston Observatories between the end of the first part of O4a and the end of O4c, including upgrades made during the commissioning break preceding O4b and during repairs performed in O4c. We describe instrumental investigations carried out at both sites designed to understand and subsequently mitigate the effect on detector sensitivity of transient glitches, narrowband spectral lines, and vibration-driven noise, among other data quality concerns. We then review the tools and procedures used to validate gravitational wave candidates and the data quality products thus supplied to searches for gravitational waves from compact binary coalescences and unmodeled transients, continuous gravitational waves, and the stochastic gravitational wave background. The efforts of the detector characterization group are essential for maintaining and improving the sensitivity and reliability of the LIGO detectors especially as observing runs lengthen and more events are detected. We conclude with prospects for LIGO detector characterization activities in future observing runs.

Keywords: LIGO detector characterization, data quality, instrumental noise

1. Introduction

The fourth observing run (O4) of the LIGO-Virgo-KAGRA (LVK) collaboration was the longest and most sensitive ground-based gravitational wave observing campaign to date. Since the first direct detection of gravitational waves (GWs) from the merger of two black holes on September 14th, 2015 by the LIGO Hanford Observatory (LHO) and LIGO Livingston Observatory (LLO) [1], the catalog of compact binary coalescences has grown with each successive observing run of the Laser Interferometer Gravitational-wave Observatory (LIGO) [2–6].

During O4, the two LIGO detectors operated as part of the LVK network alongside the Virgo detector [7] and the KAGRA detector [8]. The O4 observing run was divided into three periods. The first period, O4a, spanned from May 24, 2023, 15:00:00 UTC to January 16, 2024, 16:00:00 UTC. LIGO detector characterization efforts during O4a are described in [9]. The corresponding efforts for Virgo [10] and KAGRA [11] are reported separately. Following a commissioning break of approximately three months for the LIGO detectors, the second part of the fourth observing run, O4b, began on April 10, 2024 15:00:00 UTC. By the end of O4b, a total of 390 significant detection candidates had been identified across all observing runs [6]. The end of O4b on January 28, 2025 17:00:00 UTC also marked the start of the third part of O4, O4c. During O4c, the LIGO detectors underwent a period of commissioning during which no astronomical data was taken from April 1, 2025 15:00:00 UTC to June 11, 2025 15:00:00 UTC, to make hardware changes to the vacuum system. O4c concluded on November 18, 2025 16:00:00 UTC.

Figures 1 and 2 show representative Amplitude Spectral Densities (ASDs) of the GW strain noise at LHO and LLO, respectively, across the three parts of O4. The GW strain, $h = \Delta L/L$, is the fractional change in the differential arm motion Differential Arm Length (DARM) ΔL relative to the nominal arm length L , and is the primary data product of each detector from which GW signals are extracted. Along with ASDs, a common measure of detector sensitivity is the Binary Neutron Star (BNS) range, defined as the sky and orientation averaged distance at which a $1.4 M_{\odot}$ – $1.4 M_{\odot}$ compact binary merger with a signal-to-noise ratio (SNR) of 8 could be detected [12]. The BNS range provides a convenient measure for quickly assessing detector performance. A detailed comparison of detector performance across O4b and O4c, including the pre- and post-commissioning period of O4c is presented in Section 2.

The LIGO detector characterization (DetChar) group is tasked with monitoring the state of the detectors, identifying and mitigating sources of noise, and providing data quality information to the astrophysical analyses that search for GW signals. This work includes on-site investigations of instrumental and environmental couplings, development and maintenance of software tools for characterizing the data, validation of candidate events, and the production of data quality products consumed by downstream search pipelines.

In general, the data from the LIGO detectors is typically Gaussian and stationary.

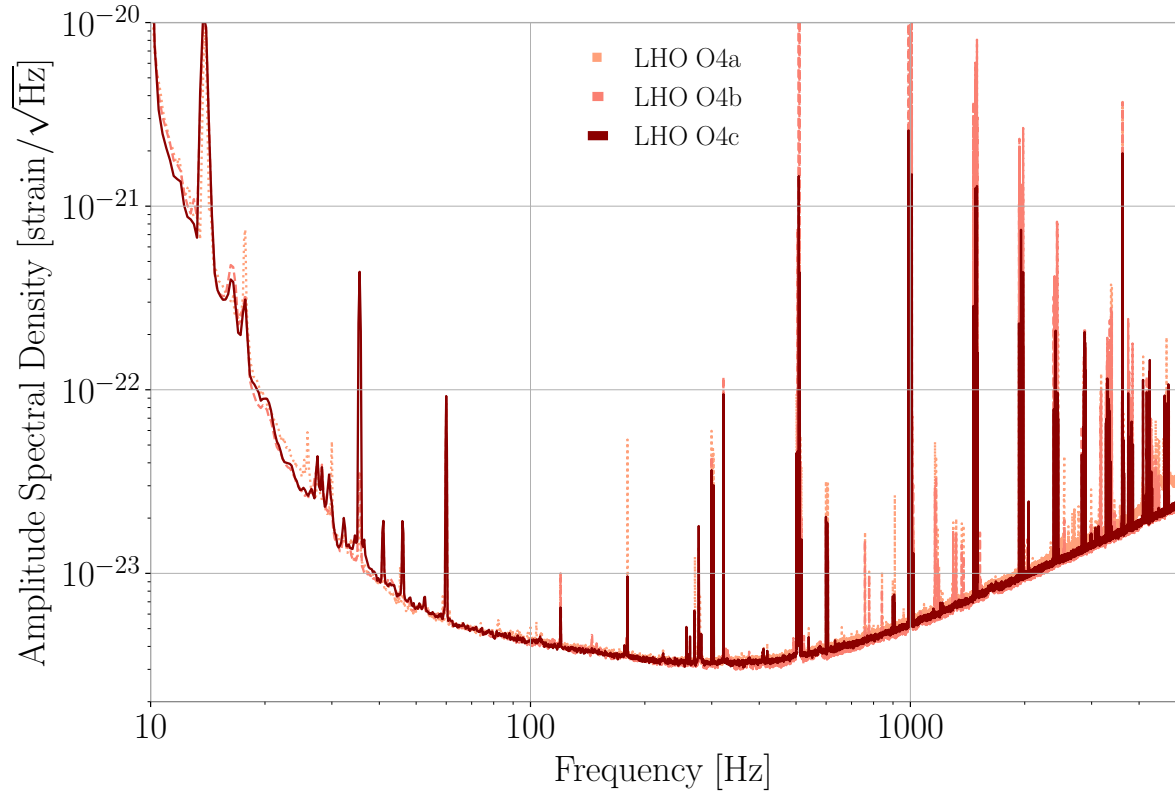


Figure 1: Representative ASDs of the GW strain noise at LHO during O4a, O4b, and O4c. The broadband sensitivity is comparable across the three epochs, with slight improvements in O4b and O4c relative to O4a in the 20–30 Hz region. The population of narrowband spectral lines and combs varies noticeably between periods, particularly above 500 Hz. These features and their variations are discussed further in Section 3.2.2 and in [13].

However, noise which couples to the detector through numerous mechanisms introduces non-Gaussian features that need to be characterized and, where possible, mitigated [14–16]. These features are commonly divided into three categories: short-duration transient artifacts (often called *glitches*), narrowband spectral lines and combs, and persistent broadband artifacts. Glitches are typically categorized by their morphology in time-frequency spectrograms [17, 18]. Narrowband spectral lines are persistent artifacts that appear as peaks in the ASD of the strain data that can often be traced back to mechanical resonances or electronics noise [13]. A comb is a set of narrowband lines that are uniformly spaced in frequency [13]. Persistent broadband artifacts elevate the noise floor across a range of frequencies. Each of these different types of noise impacts the GW searches in distinct ways. Glitches can impact searches for short-duration signals, including Compact Binary Coalescences (CBC) and unmodeled burst searches, where they can produce false triggers or reduce the significance of a genuine astrophysical candidate. Beyond affecting detection, a glitch that overlaps a signal can bias the estimation of source properties and, in turn, the downstream

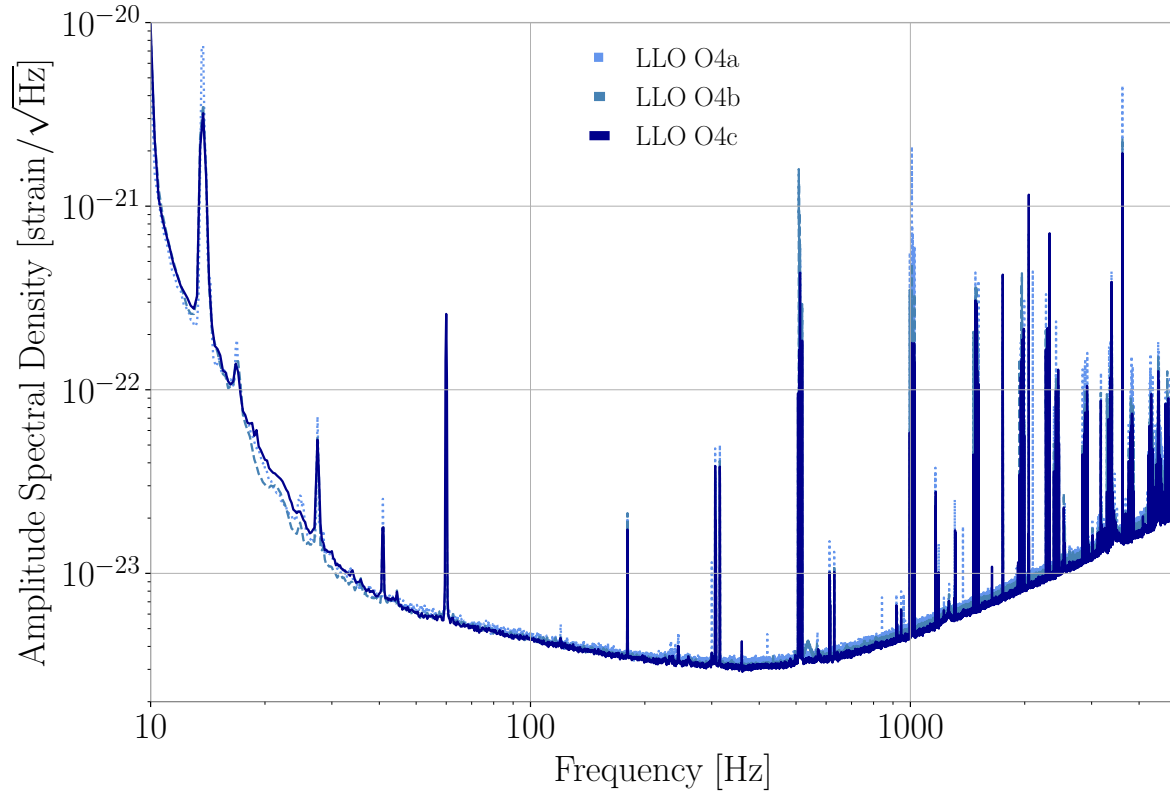


Figure 2: Representative ASDs of the strain noise at LLO during O4a, O4b, and O4c. The sensitivity improved slightly between O4a and O4b, while O4c shows a mild degradation relative to O4b in the 20–40 Hz band.

astrophysical inferences that rely on them [19–21], motivating the event validation and noise mitigation procedures described in Section 4. Narrowband spectral lines and combs limit the sensitivity of searches for Continuous Wave (CW) signals from spinning neutron stars and may introduce cross-correlated noise, affecting searches for the stochastic GW background. The presence of persistent broadband artifacts affects LIGO’s searches for both GW transients and CW and stochastic GW sources by degrading detector sensitivity across large swathes of frequency space.

The scope of this paper encompasses the LIGO commissioning and DetChar activities carried out during O4b and O4c. Section 2 summarizes changes to the detector configurations between the end of O4a and the end of O4c and the resulting effects on metrics such as detector duty cycle. The instrumental investigations carried out at both observatories to identify and mitigate sources of environmental noise during O4b and O4c are described in Section 3. Section 4 details the performance tools used to validate GW event candidates based on data quality surrounding the event time. The use of data quality products in searches for GWs of different kinds (compact binary coalescences, unmodeled bursts, continuous waves, and stochastic signals) is discussed in Section 5. We conclude in Section 6 with a summary of our results and outlooks for future DetChar work.

	O4a	O4b	O4c
LHO	139.4	149.5	145.2
LLO	149.4	159.4	146.8

Table 1: Comparison of the median BNS range (in Mpc) for the two LIGO detectors in each part of O4.

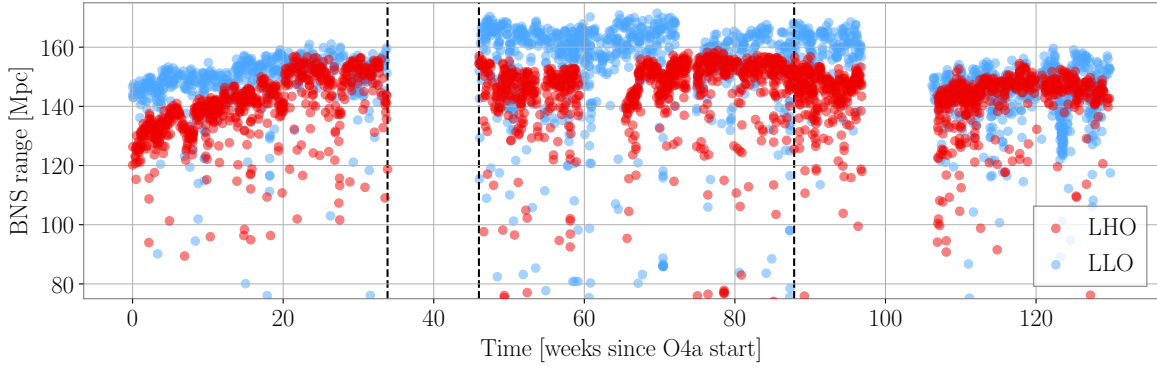


Figure 3: A comparison of the BNS range of the LIGO detectors in each part of O4. Dashed vertical lines indicate divisions between parts of the observing run. The large gaps correspond to commissioning periods where maintenance and upgrades were performed on the detectors. No astrophysical data were taken during these commissioning periods.

2. Detector Configuration & Performance in O4b & O4c

Between the end of O4a and the start of O4b, both LIGO detectors underwent a commissioning break during which a number of instrumental upgrades were made. Additional changes were implemented during the commissioning break in the middle of O4c (April 1–June 11 2025). This section describes those configuration changes and their impact on detector performance and sensitivity. The BNS range over the full duration of O4 is shown in Figure 3, and figures 4 and 5 show the range ASD at both detectors across O4b and O4c. Overall, the BNS range was slightly greater in the second part of O4 compared to the other parts of the observing run. Both detectors failed to return to their prior sensitivity following the mid-O4c commissioning break; LLO’s typical BNS range in O4c was less than its range in O4a. A comparison of the detector BNS range by observing run part is given in Table 1. The maximum BNS range achieved by any LIGO detector was 171.5 Mpc, recorded by LLO during O4b on September 23, 2024. LHO’s maximum BNS range, also achieved during O4b, was 158.9 Mpc, which was realized on November 18, 2024.

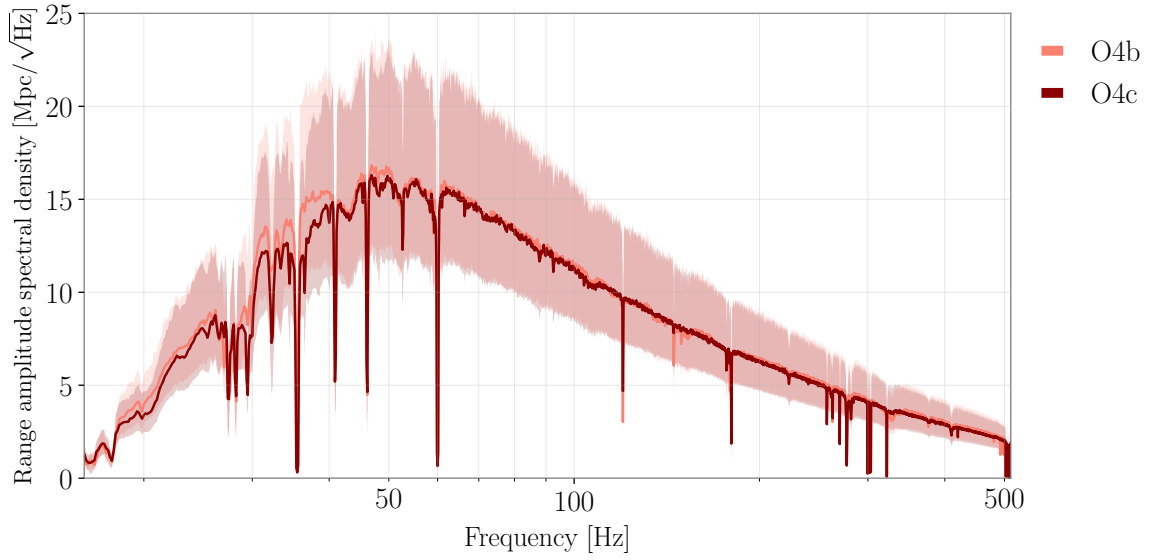


Figure 4: Range ASD for LHO during O4b and O4c. Solid lines show the median over a 24-hour period of observing time; shaded bands show the spread across short segments over the day (5th–95th percentile). Narrow dips correspond to instrumental and environmental spectral lines.

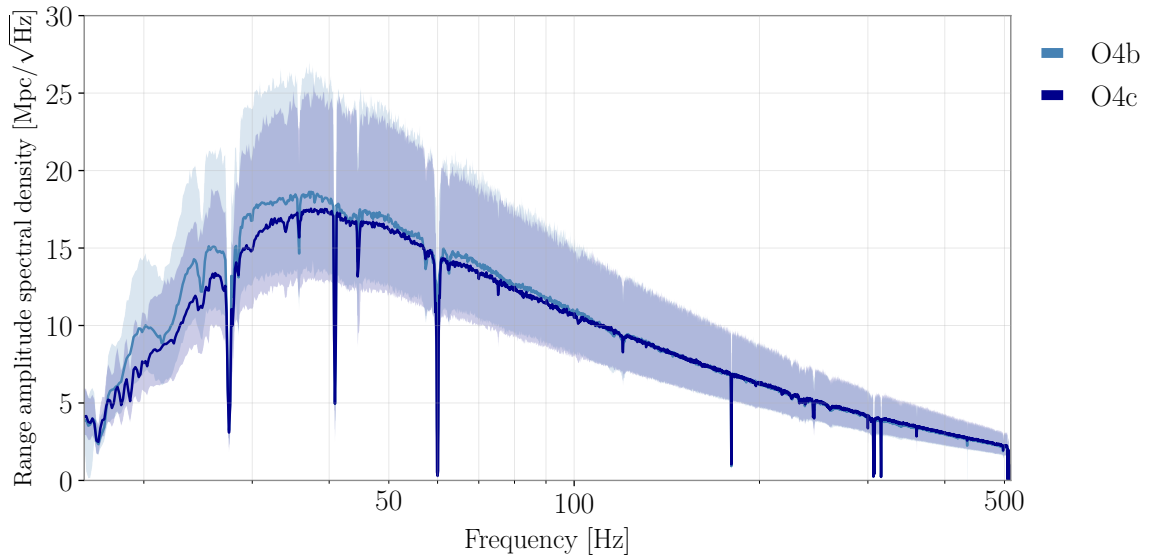


Figure 5: Range ASD for LLO during O4b and O4c.

2.1. Hardware injections

A critical component of GW candidate validation is ensuring that detected events are not caused by environmental or instrumental noise sources recorded in auxiliary channels. Several tasks described in section 4 rely on auxiliary channel information to determine whether or not a GW candidate is astrophysical in origin. Since the transfer functions between the GW strain channel and thousands of auxiliary channels are not known, there is a possibility that real astrophysical signals can sometimes appear in certain auxiliary channels due to unknown couplings between auxiliary channels and elements of the signal chain used to produce calibrated strain data. Such auxiliary channels are called “unsafe” and should not be used in the development of the vetoes described in section 5 since they can effectively allow a real GW event signal to be vetoed. To determine whether a channel responds to change in power in $h(t)$ and thus determine its “safety,” the detector characterization group performs DetChar safety injections. These injections are done using photon calibrator to inject sine-Gaussian waveforms into the $h(t)$ channels at each detector [22]. A channel is considered safe only if it shows no response to these injected signals. We then perform a statistical analysis of approximately 5,000 auxiliary channels sampled above 16 Hz using the pointy-poisson tool [23], classifying each channel as either “safe” (suitable for vetoing candidate GW events) or “unsafe.” The final list of safe channels is subsequently used in downstream data-quality analyses. DetChar hardware injections and the subsequent analysis were performed routinely during O4b at both detectors. No significant changes were observed in the safety of auxiliary channels during this time.

2.2. Commissioning changes at both detectors

As O4 was the longest observing run—to date—for the LIGO detectors, many changes, repairs, and improvements were made to both LLO and LHO during the course of the observing run. These adjustments frequently affected the duty cycle and environmental coupling of detectors. In this section, we catalog these changes, beginning with those made to both interferometers. Figure 6 depicts LLO with individual vacuum chambers labeled and shows their corresponding buildings.

2.2.1. EQ mode Earthquakes and periods of heightened microseismic noise affect the duty cycle and sensitivity of the LIGO detectors. An automated global control method called “EQ mode” was implemented in the third observing run (O3) to enhance the ability of the detector to maintain lock during earthquakes which occur more than 1000 km away from the sites [26]. In O4c, a new feature was added to EQ mode which switched arm alignment sensing and control loops to a high-bandwidth state during earthquakes [27]. This change in EQ mode controls increased the fraction of earthquakes for which the detectors were able to maintain lock, even at higher ground motion [28]. The improved performance of the detectors at maintaining lock is shown in Table 2. Using this metric as the figure of merit, the new EQ mode did not degrade

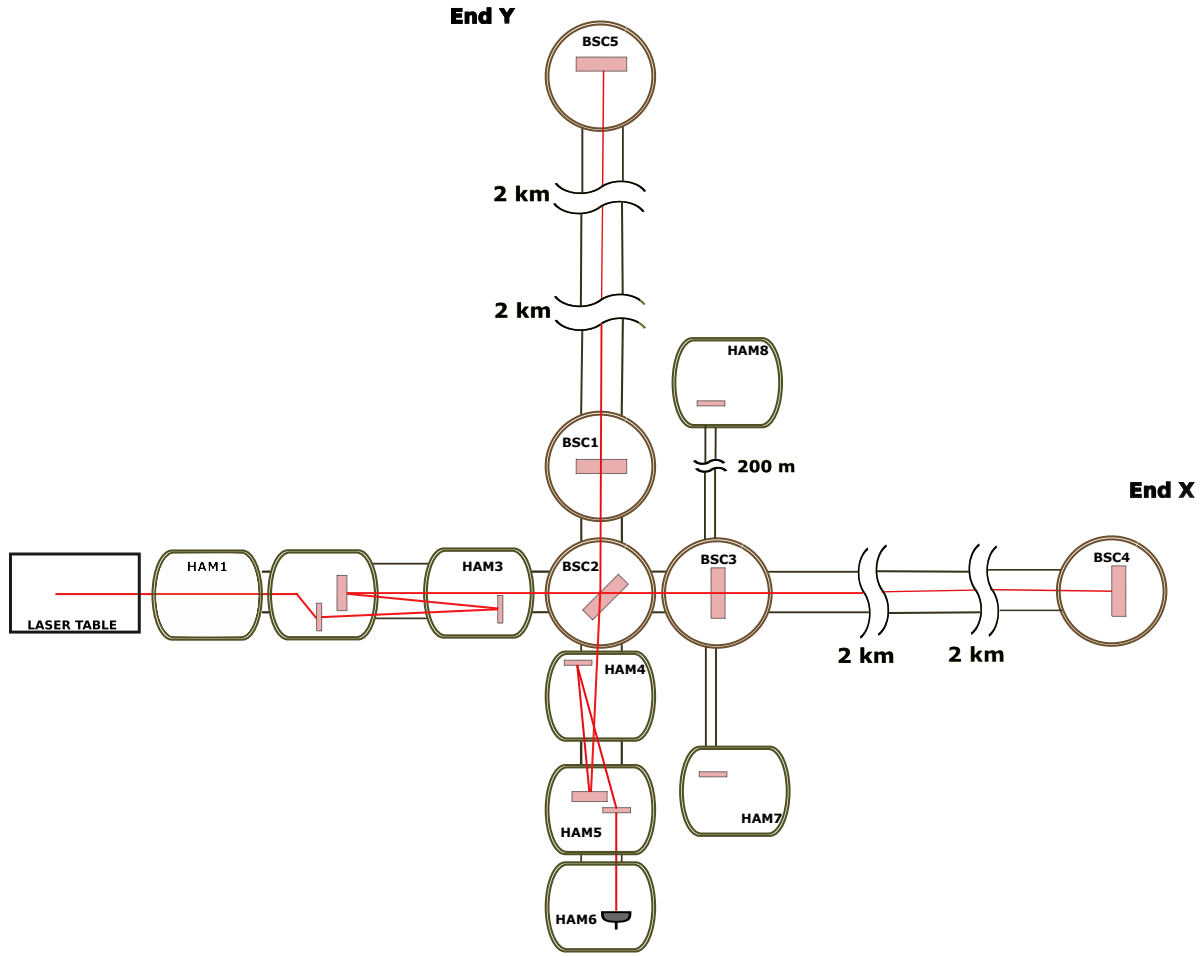


Figure 6: Schematic of LLO vacuum chambers adapted from [24]. The path of the main laser (red) off core optics (pink) is also shown. Except for the laser table, the entire system is kept under vacuum. Also shown is the filter cavity used to realize frequency-dependent squeezing [25]. Each vacuum chamber is labeled with the name used throughout this paper, though not all are discussed in this work.

duty cycle, as the fraction of earthquakes survived was about the same. Further study ahead of Intermediate Run 1, which will begin in late 2026, is ongoing to determine if the new EQ mode can be further optimized to improve lock survival during earthquakes. Maintaining lock during periods of high ground motion increases the total duty cycle of the detector, as the relocking procedure can take tens of minutes to hours depending on environmental conditions.

2.2.2. HAM-1 ISI install As of the start of O4, both LLO and LHO contained one passively-damped in-vacuum table which was susceptible to excess noise contamination,

Detector	Observation Run	Total Earthquakes	Survive Lock	Broke Lock	Lock Probability (%)
LLO	O4b	176	53	123	30.1%
	O4c	234	71	163	30.3%
LHO	O4b	193	102	91	52.8%
	O4c	203	112	91	55.2%

Table 2: Performance of LIGO detectors during earthquakes in the second and third parts of O4. Total Earthquakes refer to all earthquakes for which the detector was locked before the earthquake’s arrival. Earthquakes were identified, following the approach in [26], by coincident peaks in vertical-direction ground motion in the 30–100 mHz band in two or more ground motion witnesses in different buildings at LHO or LLO.

especially in the 10–30 Hz band [29]. This table, located in Horizontal Access Module (HAM) 1, houses interferometric sensors and single-suspension optics used to read out length and alignment degrees of freedom from the reflection and power-recycling cavity pick-off ports of the interferometer. Noise from excess motion of this table can couple to the gravitational-wave readout through the alignment controls coupling [29]. For the first two parts of O4, both detectors employed feedforward subtraction using on-table seismic sensors to suppress the noise from this table [29]. Starting April 1, 2025, this passively-damped table was replaced at both sites with a HAM internal seismic isolation (ISI) table, which uses active seismic isolation, and has much better isolation performance than passive isolation [30]. The improved seismic isolation reduced the table motion by more than a factor of 10 above 1 Hz relative to the performance of the passively-damped table.

As a part of this install, an alignment sensor that was previously installed in-air was moved onto HAM 1 at both sites. This installation included a new single-stage suspension for beam steering onto the sensor. The alignment control loops that control the power-recycling cavity alignment degree of freedom were recommissioned to use this in-vacuum sensor, which improves stability of the alignment controls during high power operation. Moving this sensor onto an isolated platform yields less risk of noise contamination from backscattering or acoustic noise. Figure 7 illustrates this improvement.

2.2.3. Recalibration of Vault Magnetometers At each site, two LEMI fluxgate magnetometers are buried in vaults outside the detector buildings, with one unit oriented parallel to the X-arm and one parallel to the Y-arm [31]. They are located away from local sources of electromagnetic interference, which allows them to measure the ambient magnetic field. This includes contributions from Schumann resonances, which enter the magnetic noise budget for Stochastic Gravitational-Wave Background (SGWB) searches [32–34].

The LEMI magnetometers were initially calibrated in 2018 and 2019 prior to the start of O3. Recalibration campaigns were conducted at both sites during O4, in May 2025 at LHO [35] and October 2025 at LLO [36]. To calibrate, wire coils were driven

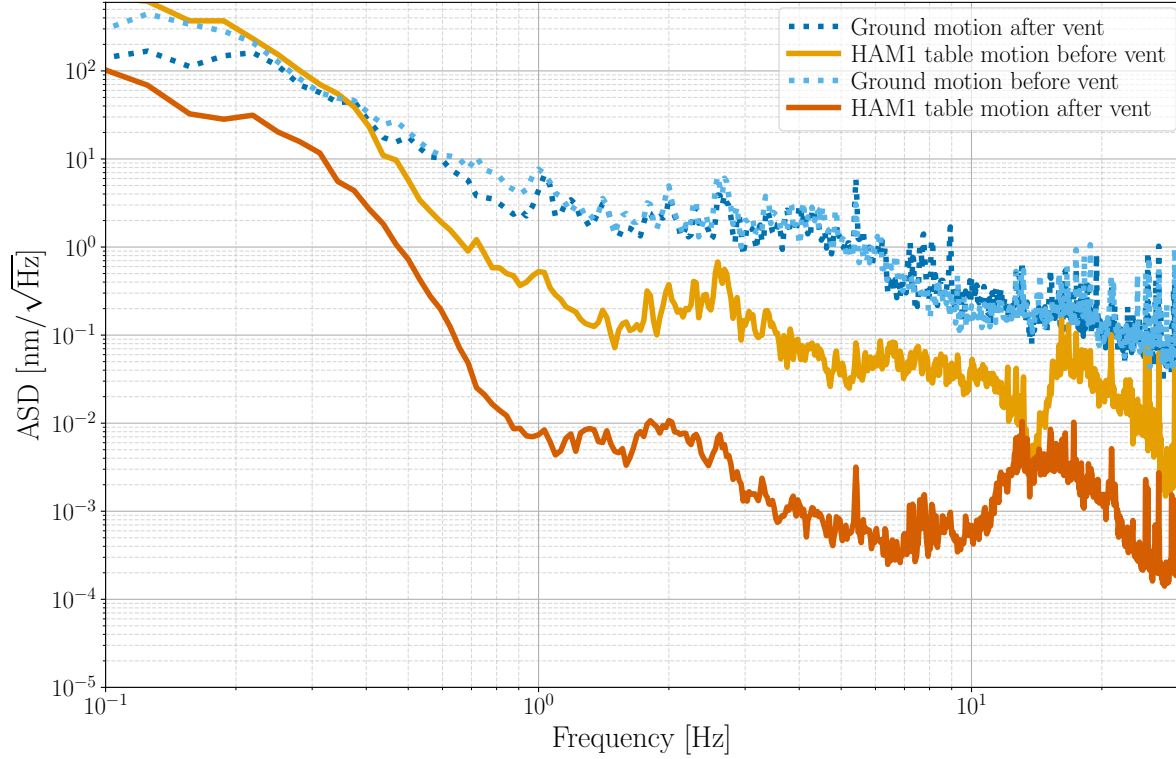


Figure 7: Vibration of the HAM 1 table-top before and after the ISI was installed at LHO.

at a known voltage and frequency to generate a magnetic field of known amplitude at the sensor location while a co-located pre-calibrated Bartington magnetometer served as a reference. The LEMI calibration factor was then derived from the ratio of the signals recorded by the two sensors. The updated calibration factors are approximately 20% lower at LHO and 70–75% lower at LLO compared to the pre-O3 values. At both sites, the discrepancy is attributed to the previous calibration campaigns not having the reference magnetometer fully isolated from the LEMI unit; in the present campaigns, the two sensors were kept completely independent.

2.3. Changes to LIGO Livingston

LLO generally operated with a larger BNS range and duty cycle compared to LHO in O4b and O4c. While a commissioning break, mainly undertaken to perform preventative maintenance to critical vacuum system components resulted in the largest gap in O4 observing time, this time was also used to reduce sources of scattered light in the detector.

2.3.1. Gate Valve 11 Repair The LIGO detectors use gate valves (GVs) to separate portions of the evacuated detector beamtube and groups of Beam Splitter Chambers (BSCs) and HAMS housing in-vacuum detector optics. A GV at LLO required repair so

it could be closed during periods of severe weather in the 2025 North Atlantic hurricane season. A mid-O4c gap spanning April 1, 2025–June 4, 2025 was used to complete this work, which restored the GV to normal operation. This gap in observing time can be seen in Figure 3.

2.3.2. Cage Baffles During the GV repair undertaken in O4c, a new set of baffles was installed at LLO near each test mass mirror. These *cage baffles* are attached to the suspension system which holds the quadruple suspension of the test masses. Positioned just in front of the mirrors, they are designed to block some of the light scattering from near the edge of each mirror, where the reflective coating does not extend all the way to the edge of the optic. The beam from the opposite test mass, 4 km away, has expanded by the time it reaches the end mirror, such that a small fraction of light falls at this diameter near the edge of the coating, making it a source of scattered light. The cage baffles were installed primarily to address the scattered light noise that limited LLO sensitivity during periods of elevated microseismic ground motion. Their resulting effect is described in Section 3.3.1.

2.3.3. L2 Split Drive In early October 2025, LLO suffered repeated lock losses, reducing the duty cycle during this period. The control system did not have enough range to hold the detector steady against transient low-frequency motion. To fix this, the DARM actuation was shared across both end test mass suspensions rather than a single one, which roughly doubled the available range and improved stability during lock acquisition [37]. Driving the additional suspension excited one of the Y-arm End Test Mass (ETMY) mechanical resonances at 13.7 Hz, which was suppressed with a notch filter [38].

2.4. Changes to LIGO Hanford

LHO commissioning activities during the second and third parts of O4 reduced broadband quantum noise sources through improved squeezing performance. However, degradation of critical equipment over the course of the observing run reduced the detector’s duty cycle.

2.4.1. Squeezer Performance in O4b and O4c Prior to O4a, the LIGO detectors were reconfigured to use frequency-dependent squeezing—rather than frequency-independent squeezing, used in O3—to simultaneously reduce radiation pressure noise (the dominant quantum noise source at low frequencies) and shot noise (the dominant quantum noise source at high frequencies) [25]. A detailed history of detector configuration changes affecting the performance of quantum squeezing in O4a is found in [29]. Continued commissioning work to improve the performance of the squeezer in O4b led to percent-level improvement in the squeezing level near 135 Hz, with improvements of half of a dB or more at higher frequencies in O4b and O4c compared to O4a. For the 1.7 kHz band

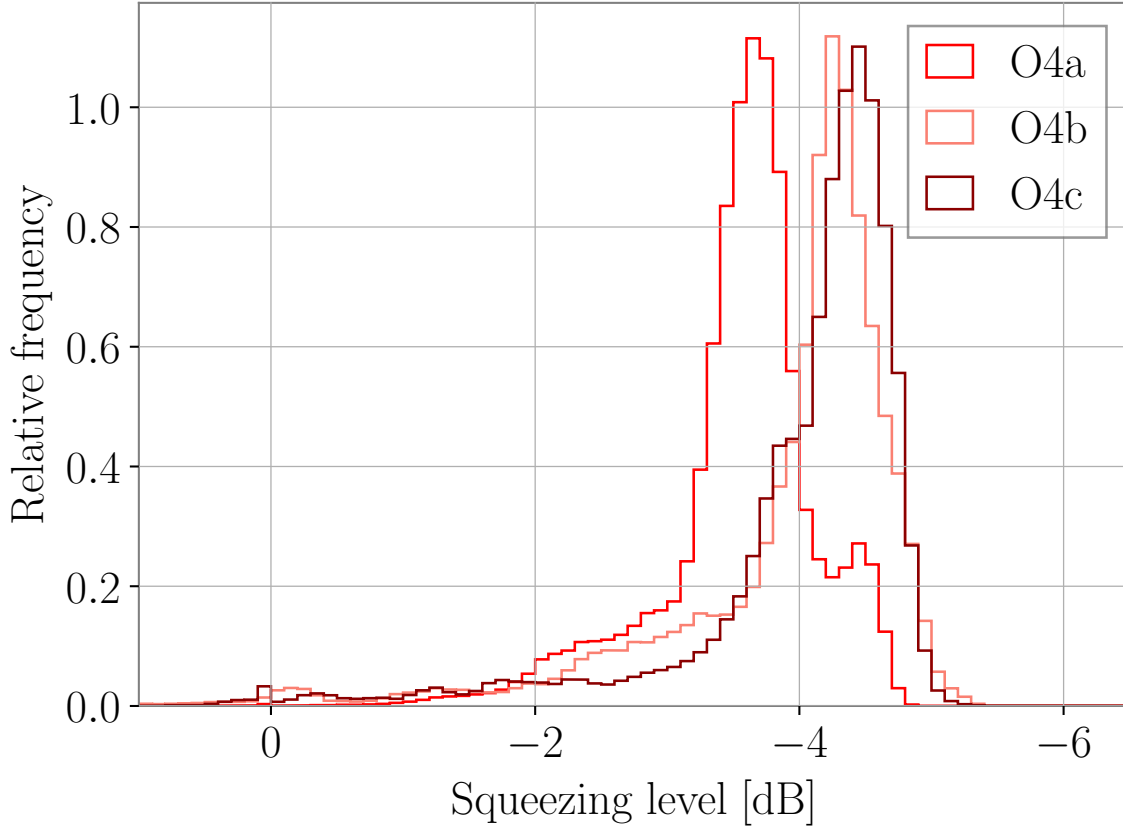


Figure 8: Quantum noise reduction from LHO’s squeezed light system in O4a (red), O4b (tan), and O4c (orange). The data shown here are measured around 1.7 kHz. In general, the performance of the squeezer at high frequencies was substantially improved following O4a, while lower frequencies saw more modest changes.

shown in Figure 8, the squeezer achieved 4.5 dB or more of squeezing for 27% (20%) of observing time in O4c (O4b), compared to just 4% in O4a.

2.4.2. Output Faraday Isolator Failure On April 23, 2024, it was observed that LHO’s output arm optical gain slightly decreased which increased scattered light noise below ~ 80 Hz [39]. Commissioning efforts identified a change in the performance of the Output Faraday Isolator (OFI) as the cause of the degradation in detector sensitivity and performance [40,41]. The OFI rejects stray light from re-entering the interferometer arms and is located on an optics table between the beamsplitter (BS) and the output port of the detector. The output arm optics were realigned to reduce noise, but this change severely affected the performance of the squeezed light system [42].

Beginning on July 12, 2024, another large change in optics alignment settings was required to keep the detector locked [43,44]. This change prompted a decision to suspend observatory operations to enter the detector and investigate the output arm in-vacuum equipment for problems. From the beginning of O4b to July 17, 2024, the duty cycle

of LHO was 54.2%, compared to its O4a duty cycle of 67.5%, in part due to instability associated with the detector in its April–July alignment.

Following replacement of failed components in the OFI polarizer [45] and wedge [46], astrophysical data-taking resumed on August 24, 2024 with a BNS range of ~ 140 Mpc, a slight decrease from its typical pre-intervention BNS range [47]. By September 6, 2024, however, LHO consistently returned to observing at ~ 160 Mpc. The impromptu break in astronomical data-taking and subsequent recovery can be seen in Figure 3 beginning about 60 weeks from the start of O4a.

2.5. Input laser system glitching

Beginning September 13, 2024, channels in the Pre-Stabilized Laser (PSL), which produces the 1064 nm light used in LIGO’s arm cavities for astrophysical observation, started witnessing glitches. The glitches occurred frequently, and were often sufficiently severe to cause the interferometer to lose lock [48]. Through September and October of 2024, the duty cycle of the detector was reduced as these glitches caused lock to be broken frequently. Analysis of locklosses with an associated glitch witnessed by frequency stabilization servo (FSS) monitors identified the non-planar ring oscillator (NPRO), the source of the 2 W in-air laser beam in the PSL, as the ultimate origin of these glitches. Between October 23 and 29, 2024, the NPRO was replaced with a spare unit used in O3 [49]. An example timeseries of one of the glitches in a channel monitoring the current sent to the NPRO piezoelectric transducer, part of the FSS subsystem, is shown in Figure 9.

The newly-installed, O3-era NPRO also suffered from glitches [50, 51] and needed crystal temperature adjustments to prevent mode hopping [48, 52]. The glitches produced by the O3-era NPRO had a different morphology in FSS data compared to the behavior shown in figure 9. Consequently, this NPRO unit was also replaced, this time with the NPRO used in the first observing run (O1) and the second observing run (O2) [53]. This unit had been refurbished by the manufacturer prior to its re-installation. The installation of the second spare NPRO on November 22, 2024 ended the frequent, lock-breaking glitches experienced at LHO.

3. Instrumental Investigations

3.1. Transient Noise

The strain data recorded by the LIGO detectors is affected by short-duration disturbances of non-astrophysical origin, commonly referred to as transient noise or glitches. These events arise from a variety of instrumental and environmental sources and exhibit a wide range of time–frequency morphologies, enabling their classification into distinct glitch groups [17].

Glitches are commonly identified using Omicron [54], which detects excess power by applying a Q-transform [55] to project the data into the time-frequency domain. This

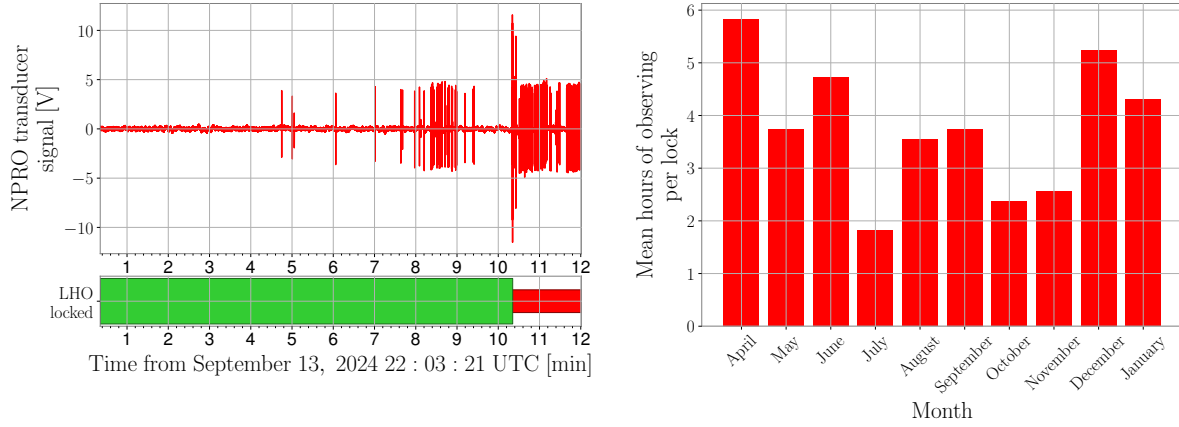


Figure 9: Left: voltage data from the FSS subsystem preceding an NPRO glitch which caused the detector to lose lock. Many smaller glitches are seen prior to the larger one which broke lock ~ 10 min from the start of the depicted time. The observatory lock status is shown by the ribbon below the plot, with the thick green ribbon representing locked times, and the thinner red ribbon corresponding to unlocked times. Comparison of timeseries of many possible witness channels in different optical cavities around LHO is what localized the glitching to the input laser system. Right: average observing-mode segment duration for each month of O4b.

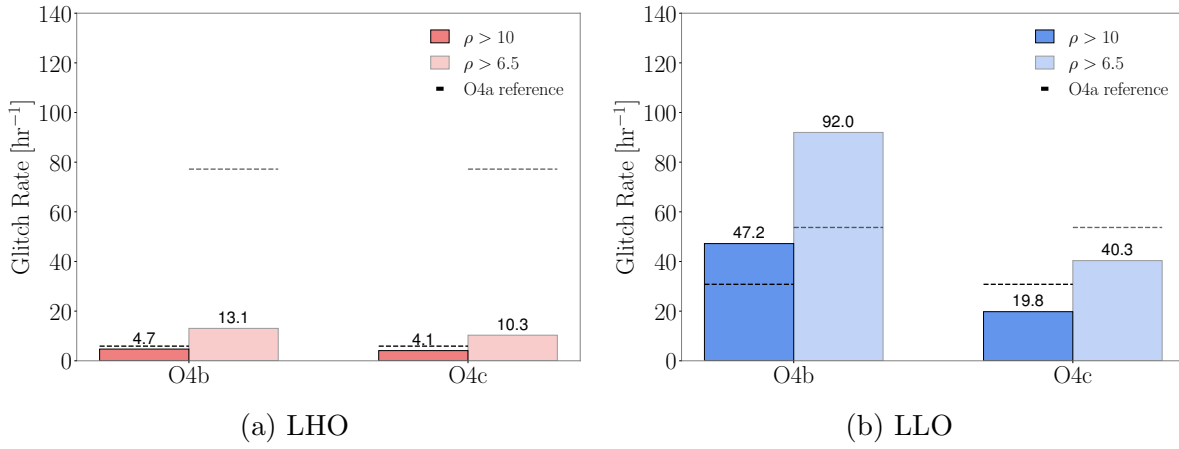


Figure 10: Glitch rate per hour during O4b and O4c for both detectors. In each panel, darker bars correspond to glitches with $\rho > 10$, while lighter bars include glitches with $\rho > 6.5$, where ρ denotes the SNR. The dashed lines indicate the corresponding O4a rates for each SNR threshold. Glitches were identified using the Omicron algorithm [54] on the “uncleaned” (GDS_CALIB_STRAIN) channel for consistency in comparisons with O4a, rather than each detector’s noise-subtracted (“cleaned”) strain channel [56].

process produces trigger-based representations characterized by central time, frequency, bandwidth, and SNR, enabling statistical studies of glitch populations and supporting their classification.

Observing run	Blip glitches per hour (LHO)
O3	1.68
O4a	1.24
O4b	1.13
O4c	0.84

Table 3: Comparison of the LHO blip glitch rate in O3 and O4 [9, 14]. Blip glitches were classified using GravitySpy [17]. Improved feedback control reduced the incidence of blip glitches in O4b and O4c.

A comparison between total glitch rates in each part of O4 is shown in Figures 10a and 10b. LHO experienced a substantial reduction in glitch rate during both O4b and O4c following the implementation of a new feedback control configuration [57] that reduced the incidence of low-SNR broadband glitches. The LHO glitch rate during O4c is the lowest observed in any detector during the advanced detector era. Blip glitches [58], a class of broadband glitches with no known auxiliary witness which mainly degrade the sensitivity of unmodeled searches, saw a decrease in rate by 50% at LHO compared to O3 following this change. Table 3 gives the rate of blip glitches in O3 and O4. At LLO, the dominant glitch population remained associated with ground motion, while the reduction observed during O4c is related to the installation of cage baffles and an ISI platform in the HAM 1 chamber [59]. The results of these changes at LLO are detailed in section 3.3.1.

A relevant change introduced in O4b is the use of the `GDS_CALIB_STRAIN_CLEAN` channel as input to Omicron, instead of the previously used `GDS_CALIB_STRAIN` channel. See Figure 11 for a comparison of the two. The cleaned channel remains calibrated and includes additional noise subtraction steps, including the removal of calibration lines, which reduce narrow spectral features and instrumental contributions that can otherwise mask transient signals in specific frequency bands. This improves the sensitivity in these regions, enabling the detection of additional glitches [60, 61].

Figure 11 shows a comparison of the monthly glitch rate in LLO obtained using the original calibrated strain and the cleaned strain channels for glitches with $\text{SNR} > 6.5$. An overall increase in the number of identified glitches is observed when using the cleaned data. This difference can be ascribed to observing more, and higher-SNR, scattering-related glitches, as these typically occur in frequency regions affected by calibration lines.

During October 2024, using the cleaned strain channel for glitch identification led to $\sim 9,000$ more glitches being detected compared to the standard calibrated strain during the same month of O4b. The effect is more pronounced for lower SNR thresholds, as weaker glitches are more susceptible to being hidden by narrow spectral features, while higher SNR events are less affected. This change will enable better studies of glitch

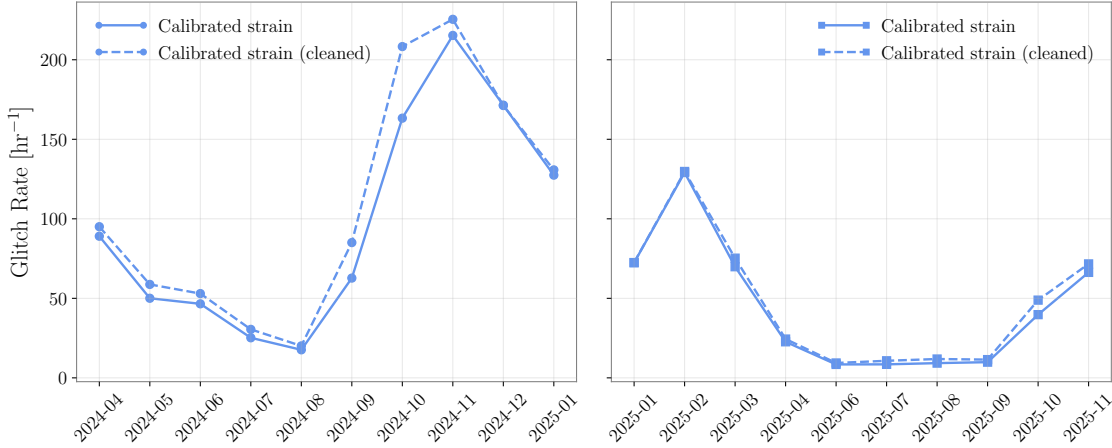


Figure 11: Monthly Omicron glitch rate at LLO for $\text{SNR} > 6.5$ comparing results obtained using the standard calibrated GW strain and the cleaned GW strain data. The cleaned data show a higher number of identified glitches, particularly during O4b, indicating that an additional population of low-amplitude glitches is recovered after the removal of narrow spectral features associated with calibration lines. The effect is less pronounced in O4c because the ambient low-frequency ground motion which produces scattering glitches was lower in amplitude during most months [62].

populations, morphologies, and correlations with auxiliary witness sensors.

3.2. LIGO Hanford

3.2.1. Vibration Coupling As the low-frequency sensitivity of the interferometer improved during O4, DARM noise peaks were revealed in the sub-100 Hz range. Many of these peaks were driven by vibration, though there was no coherence observed between DARM and the network of Physical and Environmental Monitoring (PEM) accelerometers placed around LHO [31]. Rather, the coupling was nonlinear, which suggested the coupling was likely from light scattering off optical or vacuum system components [62]. The investigations below address the drivers of these noise sources and their coupling mechanisms. In general, the greatest reductions in environmental noise contaminating the GW data were achieved by addressing the coupling mechanism between the detector and the noise source, rather than the source alone. One example of this, described in [9], is the post-O4a damping of baffles at End-X (EX).

Input Arm Baffles A second example of a vibrational noise source identified and mitigated in O4 was a broad peak in DARM data centered at 13 Hz which also added noise up to ~ 7 harmonics of its fundamental frequency. Targeted noise injection campaigns localized the coupling site to mode cleaner (MC) tube baffles and nozzles [63, 63, 64]. During the O4a-O4b break, the MC tube baffles were re-angled and nozzle baffles installed. These interventions mitigated the noise above 15 Hz. Remaining

noise at below 15 Hz was sufficiently mitigated by adjusting the Y-arm Input Test Mass (ITMY) compensation plate yaw [65].

HVAC units In the summer of 2024, multiple peaks in DARM appeared in the 24-40 Hz band. Several of these peaks were caused by heating, ventilation, and air conditioning (HVAC) systems scattered around the detector [66–69]. Many of these HVAC units employ variable frequency drives which allow for a range of operating frequencies for fans in the air handling units (AHUs). Noise driven by these HVAC units appeared in DARM data when their fan’s operating frequency range overlapped mechanical resonances of interferometer components. In instances where the AHU introducing the noise could be identified, the operating range of the AHU variable frequency drives was moved away from the frequencies which coupled strongly to DARM.

Beginning in July 2025, a large peak in DARM at 20 Hz with a harmonic at 40 Hz appeared sporadically for several months. It was most prominent during work hours, especially during the summer months. The noise source was identified as an HVAC unit located in a nearby building on LHO’s grounds. Despite being vibrationally isolated, it still coupled to DARM while in operation [70].

3.2.2. Lines and combs investigations Throughout O4, LHO consistently suffered more contamination from lines than LLO, and many investigations were carried out during the observing run [13]. During the second and third parts of the run, two problematic artifacts at LHO were targetted: a prolific comb with near-30 Hz spacing and various offsets seen at many harmonics; and investigations to understand the source of spurious, prolific line contamination in a ~ 100 Hz band near the suspension “violin resonances” seen intermittently. The near-30 Hz comb was observed in O3 LHO data as well, though to a lesser degree, and was also thought to be related to a near-100 Hz comb. Investigations using magnetometers strategically placed in various locations indicated the source of the near-30 Hz comb to be video cameras located in the PSL enclosure area [71]. When turning these cameras off, the near-30 Hz comb disappeared, but, unfortunately, the near-100 Hz comb was unaffected. Different cameras are used at LLO and are powered off during observing times, which explains why this comb is not observed in LLO data. Efforts are underway at LHO to mitigate the near-30 Hz comb, and further investigations are necessary to identify and mitigate the near-100 Hz comb.

The LIGO primary arm cavity mirrors that serve as test masses are suspended by fused-silica fibres which have fundamental standing-wave resonant frequencies near 500 Hz, commonly referred to as *violin modes* [72, 73]. Normally, the presence of these lines in the spectral data at their fundamental and higher harmonics is only problematic for the narrow region in frequency where these lines appear (~ 20 Hz at the fundamental frequency band). At times, there were a large number of additional lines observed in a ~ 100 Hz band near the fundamental of the violin modes [74, 75]. This contamination was not observed at LLO. Previously observed intermodulation between violin modes and loud calibration lines alone was insufficient to explain the line contamination. This

motivated studies of the differences between LHO and LLO, especially the GW readout differences [76–81]. A quadratic intermodulation model was investigated, allowing calibration lines and violin-mode peaks, or pairs of violin-mode peaks within or across harmonic regions to produce sum- and difference-frequency artifacts [82]. The effective quadratic term was estimated at two points along the readout, first from the analog photodetector current and then, after analog-to-digital conversion, in digital counts; the resulting patterns were qualitatively similar, but did not uniquely identify where the nonlinearity enters the readout path [82]. Further effort is needed to validate the model’s consistency with detector measurements and, if consistent, to determine which part of the readout contributes to the intermodulation.

3.2.3. Electronics Ground Noise Noise from electronics ground potential fluctuations was investigated in the first part of O4 where variation of current flowing from building electronics ground to true neutral earth were observed to be correlated with noise in GW strain data [9]. The coupling mechanism is thought to be fluctuations in the potential of the electronics ground system due to the variations in current across the finite resistance between electronics ground and neutral earth, which has been measured to be $\sim 2\ \Omega$ at LHO across many grounding spikes connected in parallel [83]. The same noise was observed at LLO through electrical injections onto the grounding system and variations in the electrostatic drive (ESD) bias for each test mass [84, 85]. At LHO, a change in the X-arm End Test Mass (ETMX) measured charge over the break between O4a and O4b suggested that there might be a change in electronics ground fluctuation coupling. A sweep of the ETMX ESD voltage found that the minimum in coupling had changed from an ESD bias of $\sim 150\ \text{V}$ in August 2023 [86] to $58\ \text{V}$ by June 2024. Changing the bias setting to this new minimum reduced the coupling by about a factor of 10 [87]. Similar measurements of the X-arm Input Test Mass (ITMX) and ITMY ESDs found that an ESD bias settings of $0\ \text{V}$ and $-40\ \text{V}$, respectively minimized the noise observed in the coherence between DARM and the current clamp on the electronics grounding cable [88, 89].

3.2.4. Magnetic Noise Beginning in O3, weekly magnetic field injections were made to monitor magnetic coupling to DARM at some locations in the corner station (CS) [31]. These injections were modified to include injections at the end stations in O4a, and were continued throughout O4b and O4c. Fluctuating magnetic couplings at high frequencies were observed in O4a [90], but the cause has not been identified. In contrast, O4b and O4c exhibit very little change in weekly magnetic coupling variation compared to previous runs [31].

3.2.5. Post-O4 Environmental Noise Injections The coupling of the LIGO detectors to their environment is quantified through periodic injections of acoustic, magnetic, and vibration noise. Detailed knowledge of the environmental coupling is used both for noise investigations and event validation [16, 90, 91]. Noise injection and coupling

function computation procedures are detailed in [31, 92]. Prior to O4a, environmental noise injection campaigns were performed at both observatories [93, 94]. PEM injections were again performed at LHO to compare the pre-run coupling to that measured at the conclusion of O4c [95].

Compared to the pre-O4 coupling shown in Figure 12, the total vibrational coupling of LHO decreased. The post-O4 measured vibrational coupling is shown in Figure 13. The most dramatic changes in EX vibrational coupling came from the EX cryobaffle damping performed before O4b [9, 65]. While vibrational coupling was reduced in the CS with changes to MC tube baffling, a worrying source of vibrational coupling, is jitter of the input laser beam [29], which is not well-described by the current linear coupling function calculation procedure, although study of other methods, like the one described in [96], is ongoing. Following the conclusion of O4c, a jitter attenuation cavity has been constructed in LHO's input arm to mitigate jitter coupling.

During the November 2025 post-run PEM injection study, we approached saturation of the accelerometers and magnetometers more closely than in past injection campaigns in order to lower the upper limits. The final post-run magnetic injections were done with more narrow bands while still covering the full range of frequencies of interest to avoid saturation. This change to noise injection procedure is most visible when comparing the magnetic injections shown in figure 14, especially at higher frequencies, to the 2025 injections in figure 15.

3.3. LIGO Livingston

3.3.1. Scattered-Light Noise Noise created by light scattering off vibrating surfaces can appear as glitches in the data, especially in the 10-40 Hz regime [59, 62]. Low frequency glitches constituted about 88% of all Omicron triggers during O4a and O4b and about 75% in O4c. Most of these glitches are due to scattered-light noise. There were two populations of scattered-light glitches: one having high SNR ($\text{SNR} > 20$) and another having low SNR ($\text{SNR} < 20$). The high SNR glitches appeared anywhere between 10-40 Hz depending on the magnitude of the microseismic ground motion. The low SNR glitches appeared only around 20 Hz and they were primarily modulated by the high frequency (10-30 Hz) vertical ground motion at the CS [59]. The instrumental investigations to better understand these two groups of glitches are discussed in the following subsections.

High SNR glitches These glitches are likely caused by light scattering off vibrating surfaces. In the first part of O4, it was found that the scattering surface(s) causing these glitches did not have any sharp resonances at a particular frequency; instead, the rate of high SNR scattering glitches very closely followed ground motion between 0.1-1.0 Hz at LLO [9]. The component of microseismic ground motion at the CS in the direction along the X-arm had the highest correlation to the scattered-light noise [59]. To find the source of the noise, sinusoidal motion was injected at microseismic frequencies at different

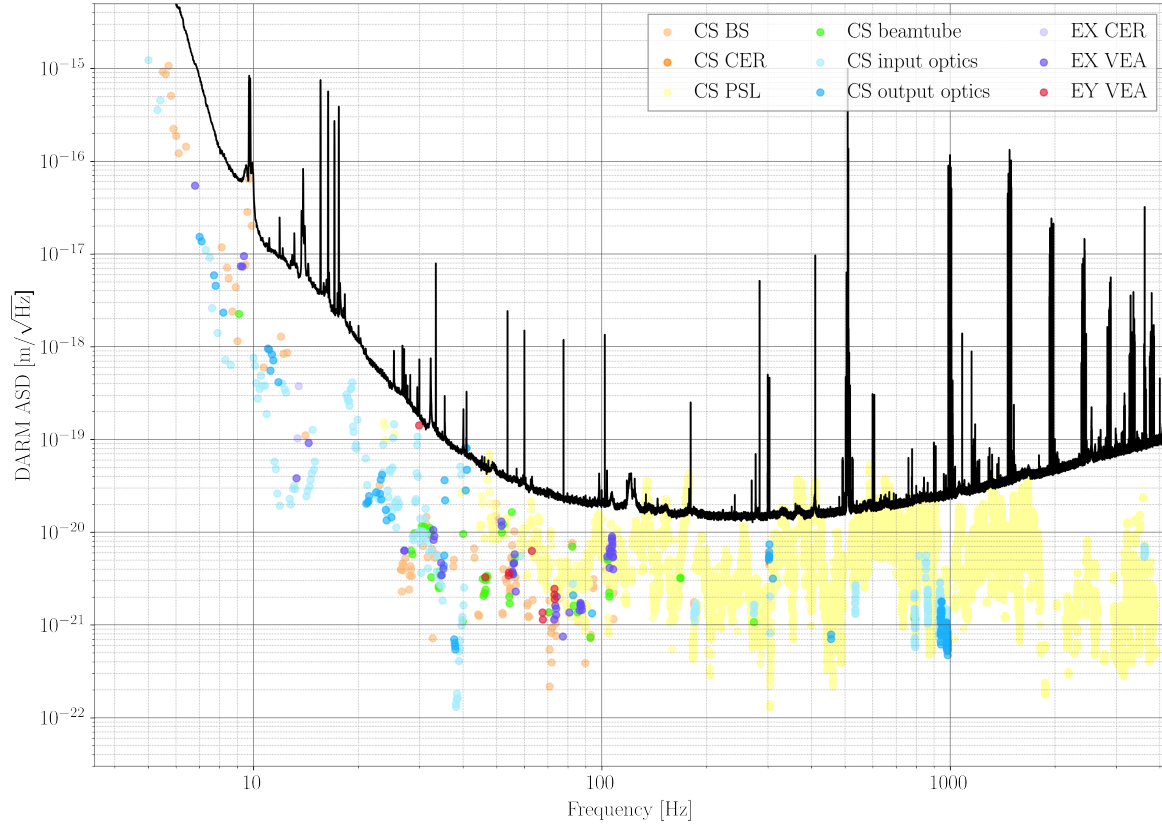


Figure 12: Results of the pre-O4 PEM vibration noise injection campaign at LHO. PEM sensors, of which there are ~ 100 at each observatory, are grouped by location, denoted by different colors. The location with the most severe coupling in each frequency bin is plotted. Sensors locations are generalized by their location in the CS Laser and Vacuum Equipment Area (LVEA) or their location in endstation vacuum and enclosure areas (VEAs) or corner or endstation controls and electronics rooms (CERs). Complete sensor coupling details may be found at [24]. Plotted points indicate frequency bins where a loud response was seen in DARM when noise was injected following the procedure in [31]. The black trace is a representative DARM ASD taken at a time close to when PEM injections were performed, but when no environmental noise was added. The circles indicate the DARM trace expected for normal levels of vibrational noise witnessed by PEM accelerometers and microphones. Vibrational coupling is largely driven by linear and nonlinear vibrational coupling to the input laser beam witnessed by CS PSL sensors. Frequencies where the prediction for DARM exceeds the observed value of DARM are within the acceptable factor of 2 uncertainty on the true value of sensor coupling to DARM [31]. While this is deleterious for noise hunting, it does not affect the performance of event validation tasks based on PEM sensor data in O4 [90].

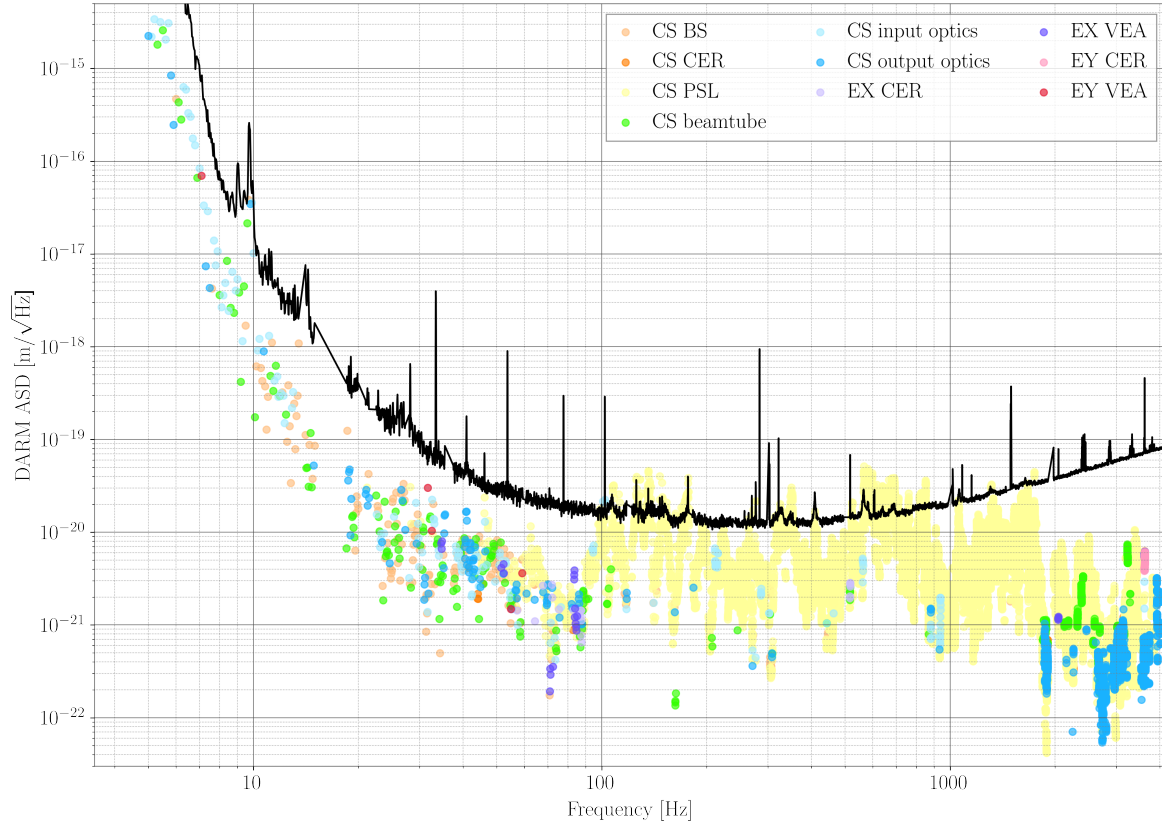


Figure 13: Results of the post-O4 acoustic noise injection campaign. The plotting style is the same as in figure 12. Baffle adjustments overall reduced the vibrational coupling, while jitter remained troublesome. The DARM trace was taken near the end of O4c.

stages of the seismic isolation and suspension system at all test mass locations and at the beam splitter. *Scatter shelves*, broadband high amplitude noise at low frequency, were observed when noise was injected at the beam splitter hydraulic external pre-isolator system (HEPI) [97], as shown in Figure 16. However, this same behavior was not seen when injecting the same amount of motion in the second stage of the beam splitter ISI platform. This suggests the elliptical baffles, because they are attached to the beam splitter HEPI but not the ISI, may be one interferometer component which scatters light that is reintroduced as noise. However, the height of these scatter shelves were almost 100 times greater in amplitude than the scatter shelves observed during elevated microseismic activity in O4. This indicates that the elliptical baffles had much more stray light illuminating them during noise injection tests than the amount of stray light which produced scatter shelves in O4. Even if a small fraction of this scattered beam gets reflected back by some other surface in the vicinity, then it can create noise similar to that seen in O4. When similar injections were performed at the test mass HEPIs/ISIs, we did not see any clear scatter shelf.

Following the installation of the cage baffles at LLO, as described in Section 2.3.2, the rate of these high SNR glitches dropped significantly. The remaining glitches also

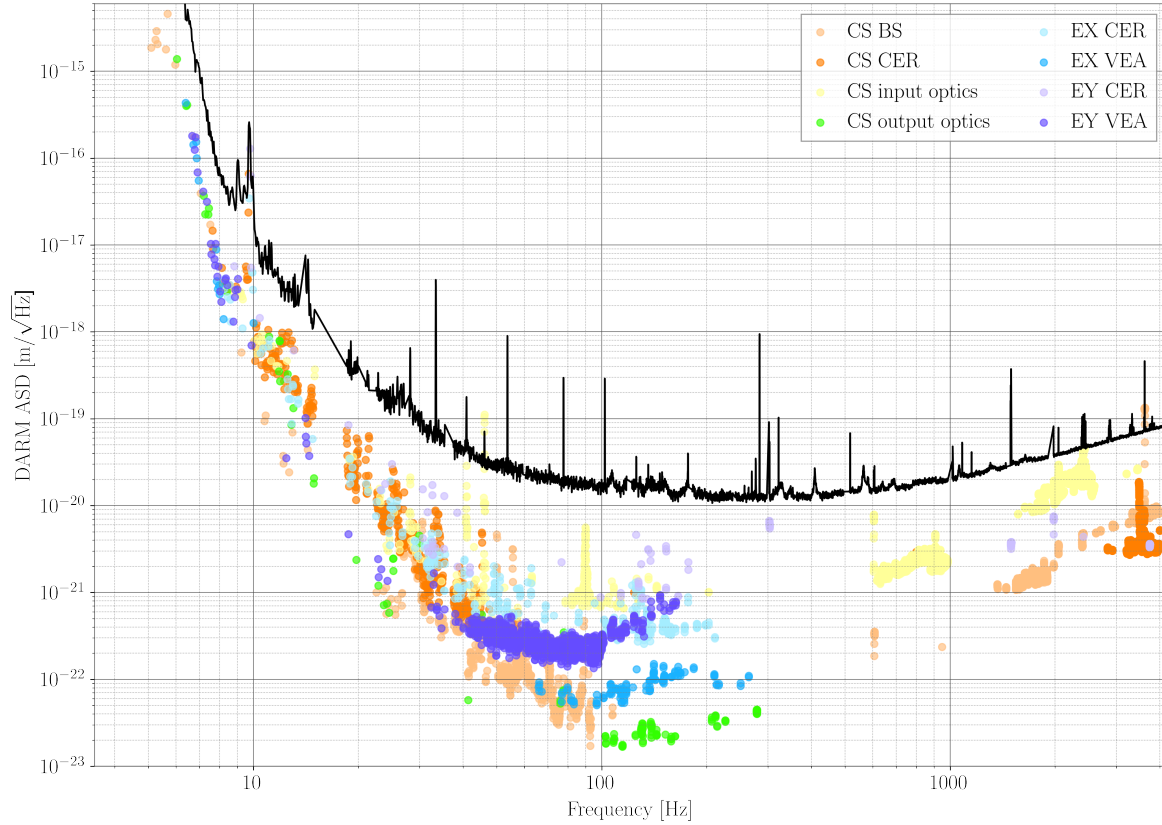


Figure 14: Results of the pre-O4 PEM magnetic noise injection campaign. The plotting style is the same as in Figure 12. Magnetic coupling in the input arm and BS areas in O4a is motivated by how close the estimated ambient value is at high frequencies to the true value of DARM and the weekly fluctuations noted in [9].

appear at lower SNR, consistent with the baffles reducing the amount of light falling on the scattering surface [98].

Low SNR glitches The rate of these low SNR glitches is primarily modulated by higher frequency (10-30 Hz) vertical ground motion at the CS [99]. Motion from the HAM 1 vacuum chamber had a strong coupling to DARM [100]. During the mid-O4c commissioning break, an ISI platform was installed in the HAM 1 chamber to provide better seismic isolation to the optics table in this chamber. After installing the ISI, which is described in Section 2.2.2, the low SNR glitches did not appear again given the same amount of ground motion [59].

3.3.2. Binary Neutron Star range noise investigations

Quasi-periodic BNS range oscillations The BNS range oscillations, described in [9], continued to occur throughout O4b and O4c. These oscillations occurred with a period of approximately 30 minutes, producing range variations between 5-15 Mpc, and are

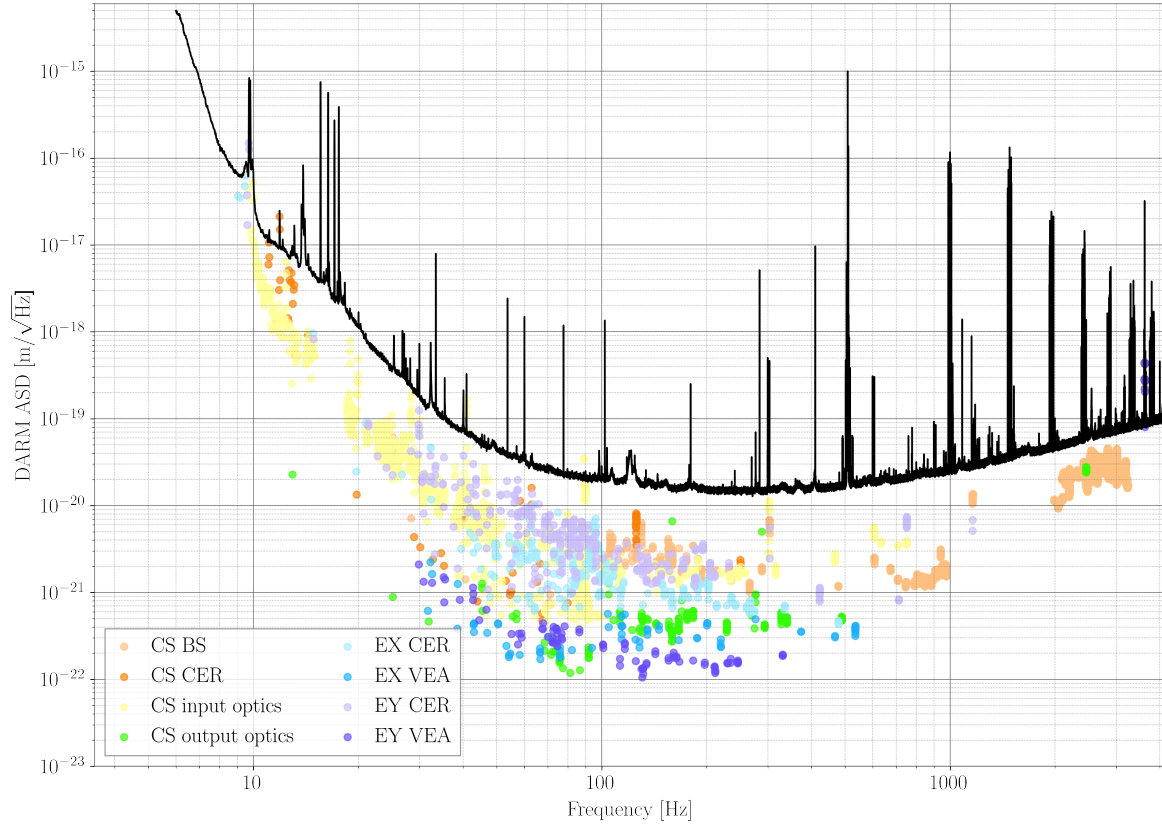


Figure 15: Results of the post-O4 PEM magnetic noise injection campaign. The plotting style is the same as in Figure 12. Improvements in both the measurement procedure and postprocessing of results permitted better understanding of the magnetic coupling following O4c.

associated with broadband excess noise in the strain data between 30-50 Hz. The oscillations were not present at all times, but when active could persist for all or part of a day, as shown in Figure 17. Although O4a investigations potentially implicated temperature effects at the EX, the coupling mechanism was not identified. Further investigations during O4b and O4c sought to identify the source more precisely.

A bakeout blanket was placed on the EX manifold, which coincided with a reduction in the frequency and amplitude of the oscillations [101], though this has not been formally confirmed as a causal intervention. Low frequency tapping tests were also performed near the EX cryopump, which produced strain noise with a character similar to the half-hour oscillations [102]. More investigation is needed into whether or not a mechanical coupling at this location plays a role.

Periodic variation in voltage monitors which monitor the EX ESDs was found to correlate with range oscillations, although a positive identification could not be conclusively established due to potential correlations in these channels with the ambient temperature [103, 104].

LASSO regression analyses [105] applied to multiple periods of O4b LLO data

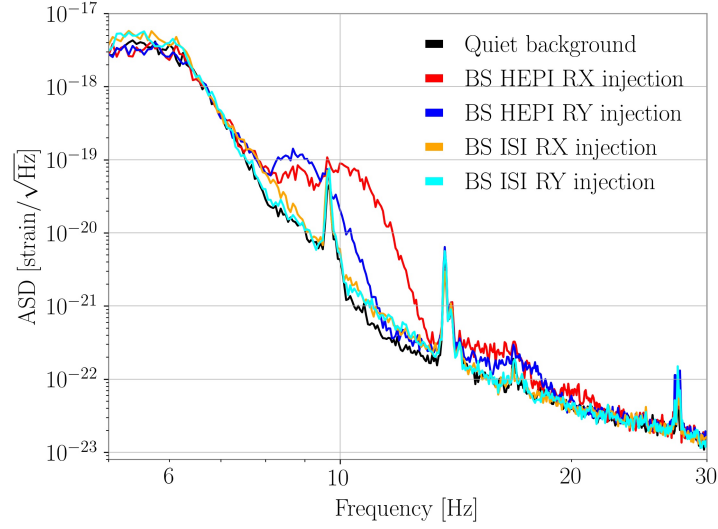


Figure 16: Scatter shelves (broadband high amplitude noise in between 8-15 Hz) produced in DARM by the 0.2 Hz BS injections compared to a reference time with no scatter shelf present. Each spectra was computed from 180 s of data. Here, “R” denotes rotation about the labeled axis. Similar amount of motion was injected at longitudinal DOFs of BS HEPIs as well but no scatter shelves were observed during those injections.

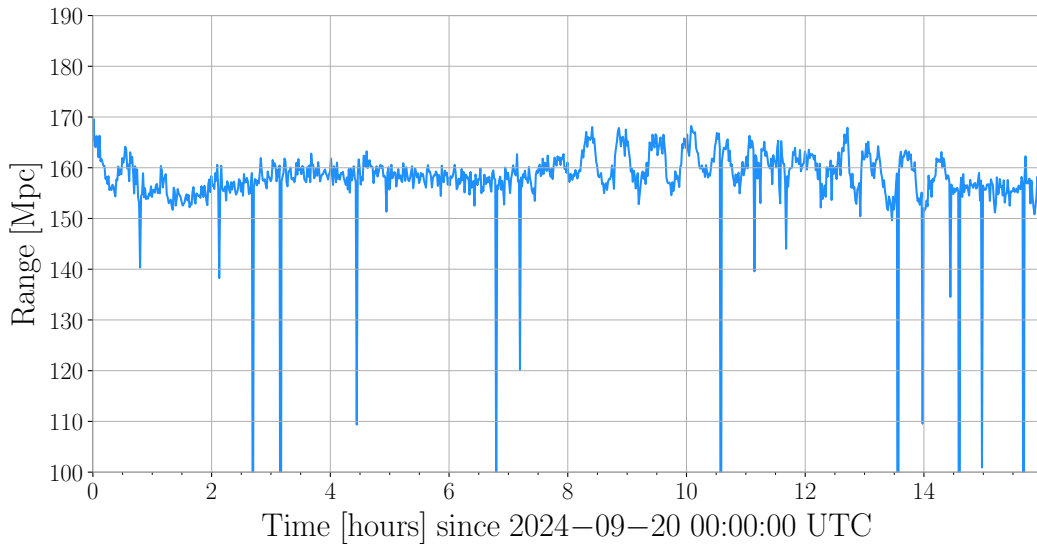


Figure 17: BNS range oscillations over a stretch of 16 hours. 30-minute oscillations are present from roughly 8 UTC to 14 UTC on this day.

consistently identified the EX as the location of subsystems most strongly correlated with the strain noise. Many auxiliary channels across several subsystems exhibited oscillatory behavior consistent with the range variations, including temperature sensors, humidity sensors, PEM sensors, and channels associated with the Arm Length Stabilization (ALS) system. Among the ALS channels flagged was `L1:ALS-X_REFL_CTRL_OUTPUT`, which is electronics noise. However, this electronics noise does drive the ALS voltage-controlled oscillators, which are a part of keeping the ALS phase-locked loops locked even when the ALS is not in use. As a test, the phase-locked loops were unlocked at both end stations while the interferometer operated in a low-noise state beginning in February 2025. In this configuration, the BNS range oscillations persisted [106]. The channels identified by LASSO spanned several subsystems and varied from day to day, reflecting the fundamental challenge that when many auxiliary channels exhibit similar oscillatory behavior, regression alone cannot determine which, if any, represent a causal driver.

To address this limitation that regression methods cannot resolve the temporal ordering of correlations, a time-lag cross-correlation method was developed to test whether changes in auxiliary channels precede or follow changes in the strain noise [107]. Preliminary results suggest that channels belonging to the facilities management and control subsystem at EX, in particular the AHU heat switch, most frequently led the strain response, pointing toward a possible environmental driver. Ongoing work aims to extend this analysis to channels at the CS and End-Y (EY).

12 hour cycle BNS range drops During O4b, LLO exhibited periodic drops in the BNS range with a cycle of approximately twelve hours, during which the range decreased by up to 20 Mpc relative to its nominal value before recovering. An example of this behavior is shown in figure 18. These drops were associated with low frequency broadband strain noise. An on/off test of the squeezer was performed, and it was concluded that the squeezer was not the source of the noise [108]. Coherence was observed between the strain data and a photodiode monitoring transmission through the EX test mass during these low range periods [109]. This suggested that beam clipping or misalignment at the end test mass may have contributed to the broadband noise responsible for the drops. After the mid-O4c commissioning break, this coherence was no longer observed [110].

Correlations between the range drops and environmental temperature sensors at the corner station were also investigated as a potential source of the noise. Initial observations noted a potential correlation between the BNS range and the channel `L0:FMC-CS_LVEA_AVTEMP`, which monitors the average temperature of the Laser and Vacuum Equipment Area (LVEA) in the CS [111]. This motivated an investigation into whether localized temperature fluctuations in the LVEA were inducing thermal lenses on the input test masses that could degrade the interferometer sensitivity. Subsequent analysis found that temperature fluctuations localized to the region around the BSC chambers, where the input test masses are housed, were correlated with negative thermal lenses measured by the Hartmann wavefront sensor in ITMY [112]. Further analysis

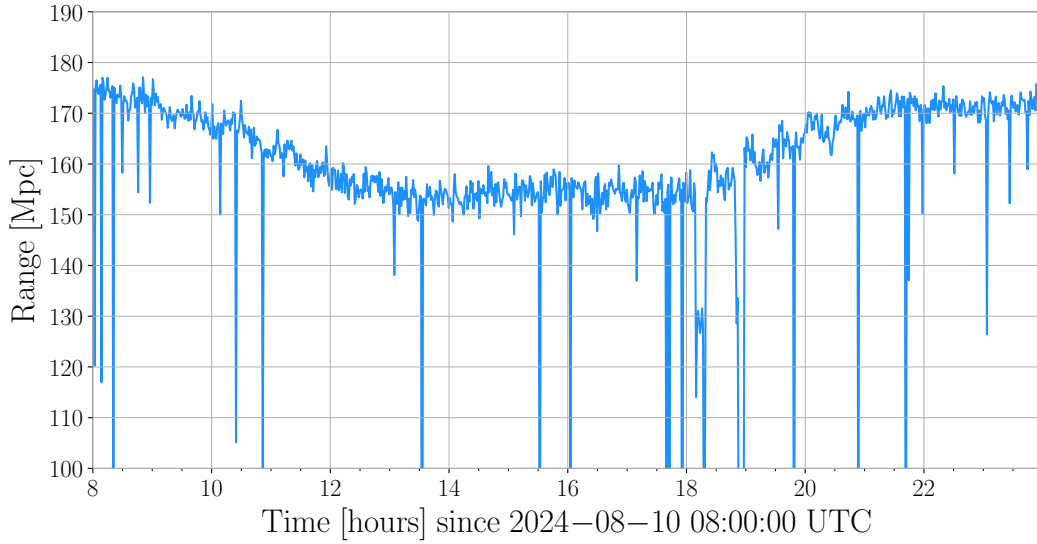


Figure 18: Example of the BNS range drop associated with low frequency broadband strain noise. The range decreased by roughly 20 Mpc during this period.

over a five-day period showed that the spherical power measured in ITMY exhibited variations with a cycle consistent with the 12-hour range drops. Variations in the DARM band-limited root mean square (rms) channels correlated with these thermal lens fluctuations were identified as a plausible mechanism by which the thermal lensing could be lowering the range. The LVEA temperature control was adjusted during this period [113–115] to make it more stable. However, the coupling mechanism was not conclusively established, and further modeling work is ongoing.

3.3.3. Wandering narrowband features during O4 In addition to the stationary spectral lines in section 3.2.2, O4 data also contained narrowband features whose central frequencies drifted over time. One significant example observed at LLO was a wandering feature near $\sim 550\text{--}640$ Hz, with a frequency excursion of order ~ 30 Hz, which was noted near the beginning of O4 and remained visible at least throughout the observing run. This wandering behavior is shown in figure 19. Related wandering line structures were also observed at higher frequencies, including features near ~ 3.4 kHz, ~ 5.0 kHz, and ~ 7.9 kHz, with some of the higher-frequency features showing inverted frequency evolution relative to the lower-frequency line [116]. Follow-up studies using high-sample-rate auxiliary data indicated that these structures were unlikely to be produced simply by aliasing from the 512 kHz to 16 kHz data stream [116]. Auxiliary channel investigations found modest correlations with environmental and interferometer channels, including an LVEA temperature probe and a local-oscillator monitoring channel associated with the 42 MHz squeezer system, suggesting that slow environmental or instrumental variations may contribute to the observed drift, although the detailed coupling mechanism remains unresolved [116].

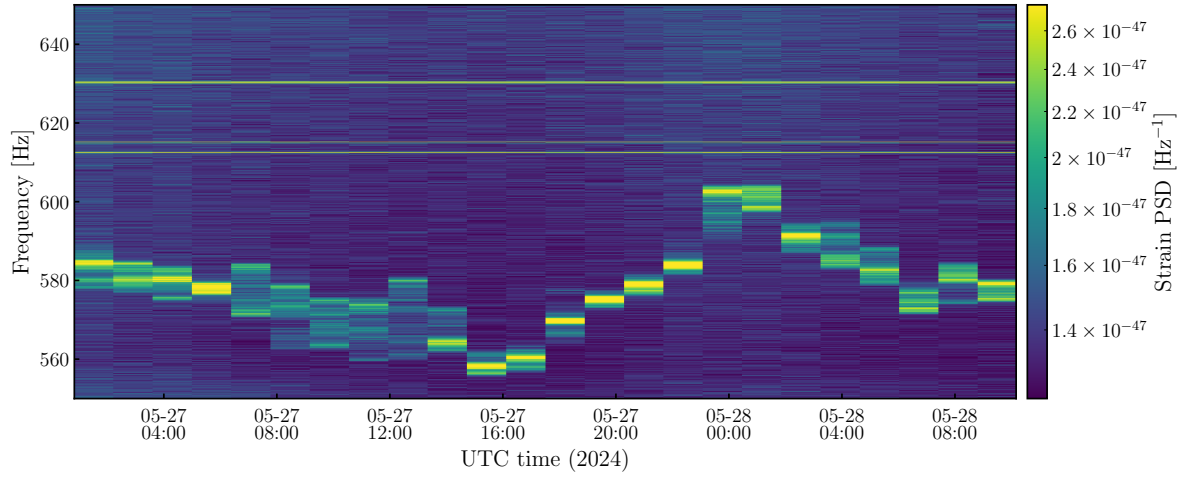


Figure 19: Spectrogram of LLO strain data over a representative continuous observing stretch during O4, showing a wandering narrowband feature near 550–640 Hz with a frequency excursion of order ~ 30 Hz.

3.4. Computational Tools

Numerous computational tools were employed in the course of these and other commissioning and noise-hunting activities described in this section as well as section 2. These tools include automated rapid online data access and visualization toolkits [117–119], analyses which identify correlations between auxiliary channel timeseries data [105, 120] or clustered noise artifacts [54] and noise/glitches in detector strain data [121–124]. Citizen-science based strain data glitch classification results were also central to many studies [17]. Also used were packages which specifically monitor the behavior of environmental witnesses [31, 125] and those which search for persistent noise artifacts [126–128]. While not all tools contributed to all investigations mentioned herein, the suite of data quality tools available to detector scientists and data analysts permits noise studies to improve the astrophysical reach of the LIGO detectors.

4. Event Validation

A central responsibility of the LIGO DetChar group is the rapid and rigorous assessment of GW candidate events. The candidate events are triggers identified by the online search pipelines PyCBC, GstLAL, MBTA, SPIIR, AFRAME, MLY and cWB as possible GWs [129–136]. The event validation (EV) process determines, for each GW candidate, whether the trigger is more likely to be astrophysical or a product of instrumental or environmental noise. This process also checks if there are artifacts that could bias analysis of an astrophysical signal. The O4b and O4c procedure of event validation was similar to what was done in O4a [9], with several targeted improvements. During O4b and O4c, 41 volunteers from the LVK collaboration assessed the detector data quality around the vicinity of GW candidates.

The EV process included verifying whether the detector was observing in a nominal configuration at the time of the candidate, recognizing noise artifacts that would bias source property estimation, checking PEM sensor outputs for any signs of environmental coupling into strain data, checking stationarity of the data, and also identifying statistical correlations with strain and auxiliary sensors. If the data quality is found to be inadequate by the DetChar experts, additional mitigation techniques - including Bayesian noise inference and transient noise subtraction - may be applied by the data analysts [137].

In the following subsections, we detail the changes in detector network, data quality report (DQR), EV infrastructure and procedure during O4b and O4c, and validation of events in these runs.

4.1. Data Quality Report

The DQR is the principal automated tool that informs event validation. For each candidate event, it runs a fixed suite of analyses around the time of the trigger — including environmental noise estimates, strain–auxiliary channel correlations, glitch predictions, excess-power and stationarity tests, and detector-range monitoring — and reports qualitative information about the state of the detectors, and for many tasks includes a label (Pass, DQ Issue, or Task Error) with a statistical estimate of noise contamination [54, 90, 123, 138–143]. The architecture of the report, including its rapid (few-minute) and offline (few-hour) tiers, is unchanged from O4a and is described in detail in [9].

In the EV workflow, the DQR’s automated flags determine which candidates require human follow-up in low latency, which is on the order of 5 minutes. Candidates for which any task reports a DQ Issue or Task Error label receive additional scrutiny from the on-shift rapid response team (RRT) while candidates with no flagged issues require no further human review in low latency. All candidates are subsequently re-examined offline, regardless of the low-latency outcome. The EV task force comprises three teams: the event validation volunteers, the noise mitigation team, and the final reviewer. The event validator, assigned on a one-week shift, assesses each active detector and records a conclusion of “Not Observing”, “No Data Quality Issues”, or “Data Quality Issues” based on the information available to the RRT as well as DQR tasks, the results of some of which take hours to produce a result. The final result of EV for a candidate which is not retracted is a recommendation, if necessary, for regions of time-frequency space that should be avoided for accurate parameter estimation.

The principal upgrade to the DQR framework for O4b and O4c was the replacement of the single statistical threshold used to flag data quality issues with per-task thresholds. Each task was assigned its own numerical threshold, calibrated using the false- and true-alarm statistics observed in O4a; this improved the overall true-alarm rate of the report while reducing spurious flags. In O4a, the DQR was based entirely on the DQRBUILD [144] package; in O4b, additional tools in the VirgoDQR [145] package

were incorporated.

4.2. Expanded detector network support

One major change from O4a was the inclusion of Virgo data in O4b and O4c. The DQR infrastructure was extended to consistently handle the three detector network configuration. DQR tasks have been successfully run on Virgo data surrounding GW candidates and their results have been used in GW candidate assessments.

After the Noto earthquake on January 1, 2024, KAGRA rejoined O4c on June 11, 2025. KAGRA data was not used during parameter estimation, and hence it was not used for validation of candidates during the remainder of O4c. However, KAGRA data was ingested and analyzed by DQR tasks in O4c.

4.3. Noise Mitigation

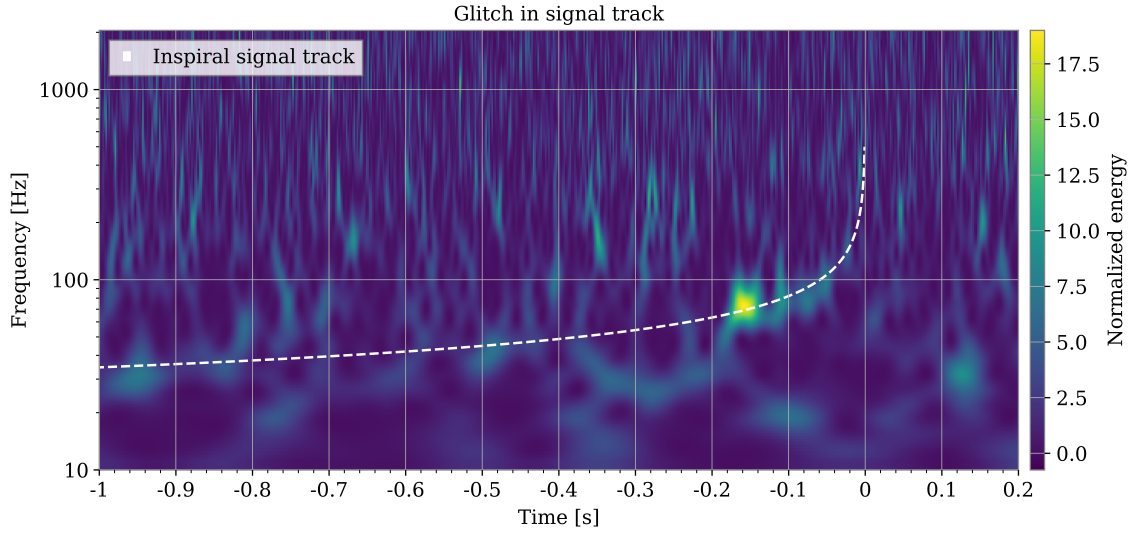
The noise mitigation team analyzes the time-frequency window recommendation for parameter estimation. In O4b, the mitigation team searched for non-Gaussian artifacts within this recommendation window and performed noise subtraction on found glitches. In O4c, the noise mitigation team less frequently required noise subtraction based on the recommendations in [146]: glitches that overlap the signal in time and frequency were removed, but any other glitches with $\text{SNR} < 50$ were kept as is. Noise subtraction was performed by the BAYESWAVE algorithm [4, 137, 147].

If the noise is extended in time or frequency, then the noise mitigation team usually recommends restricting the parameter estimation analysis window. Once the noise subtracted data frames are checked for Gaussianity, the frames and recommendations are sent to analysis groups through CBCFLOW [148] to perform parameter estimation and population studies.

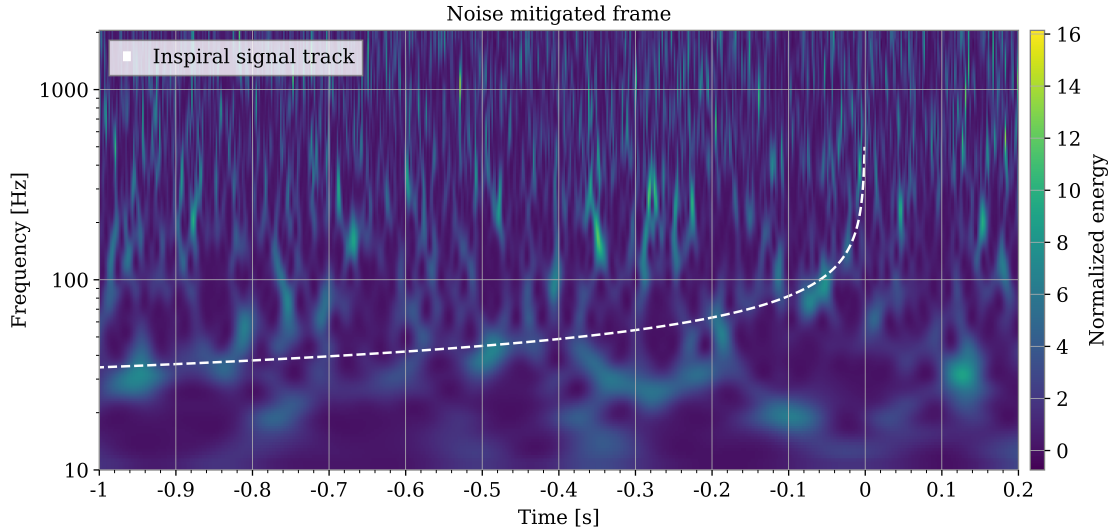
4.4. Validation of O4b and O4c events found by online search pipelines

In O4b, 114 significant detection candidates were shared as public alerts, out of which 9 were retracted. The majority of these retractions were issued due to concerns from search pipelines, rather than data quality. However, three retractions were motivated by scattered light noise overlapping the time-frequency track of the inspiral in O4b [149–151] and one retraction [152] was informed by environmental noise transients witnessed by PEM accelerometers coupling to strain [153]. In each of these cases, the retraction decision was reached by considering both data quality and search pipeline performance concerns.

Of the unretracted candidate events, 25 required glitch subtraction in O4b [6, 56]. It was seen that not all glitches affect the parameter estimation, as recommended in [146]. Thus the need for noise mitigation was reduced in O4c. An example of a glitch mitigated frame is shown in figure 20 for the event GW240520.213616 [6].



(a) Frame with glitch in signal track



(b) Noise mitigated frame

Figure 20: Example of glitch mitigation performed during O4b for the event GW240520_213616.

For low frequency extended noises, a low cutoff of 30 Hz was recommended for 10 event candidates in either LLO or Virgo, and 40 Hz for one event candidate in LLO. There was one event candidate where the low frequency was set at 70 Hz and one at 50 Hz in Virgo due to the presence of huge numbers of glitches in the low frequency region. In three instances where the transient noise environment at Virgo was sufficiently severe that glitch mitigation via noise subtraction was deemed intractable, the Virgo data were excluded from parameter estimation analysis for the affected event candidate.

5. Data Quality for Astrophysical Searches

DetChar plays a central role in mitigating instrumental and environmental noise sources that can affect astrophysical searches. Although direct mitigation at the instrument level is always preferred, it is not always possible to eliminate all sources of noise. As a result, residual non-Gaussian and non-stationary features persist in the data. To address this, the DetChar group develops and provides a range of data quality (DQ) products that are applied across astrophysical searches to reduce the impact of these artifacts.

In this section, we discuss the data quality products produced by the LIGO DetChar group which are shared across astrophysical analyses, as well as DQ measures that are tailored to specific searches depending on the nature of the source and the analysis method. Similar, but not identical DQ products, are used in the analysis of Virgo data [10, 154]. These search specific applications are generally organized into four main categories, which also reflect the major classes of GW sources: CBCs, burst, CW, and SGWB.

5.1. Data quality products for all searches

As in previous observing runs, periods of data considered unusable due to severe DQ issues are removed prior to analysis through the use of DQ flags. These flags, which differ among search groups, are distributed via the Gravitational Wave Open Science Center [155].

Category 1 (CAT1) flags define time segments that should be excluded before any astrophysical search is performed. They are broadly consistent across search groups, but small differences exist between search-specific CAT1 flags due to the different analysis methods. Throughout O4, CAT1 flags continued to be applied sparingly, reflecting improved robustness of search pipelines to DQ issues. Rather than aggressively removing data, periods with partial usability, such as those affected only in limited frequency bands, were often retained. Nevertheless, segments with severe glitches, calibration issues, or detector configuration issues were excluded. The overall analysis time removal due to CAT1 flags remained below $\sim 0.1\%$ for each LIGO interferometer. CAT1 flags in O4b and O4c included, but were not limited to:

- Periods of incorrect line subtraction, particularly at the beginning of lock stretches.
- Periods of parametric instability [156] leading to large amplitude excitations before loss of lock.
- Issues in the squeezing control system impacting the detector sensitivity.
- Severe ring-ups of violin modes, affecting calibration and data stationarity.
- Periods of incorrect observing mode definition or missing $h(t)$ data.

Category 2 (CAT2) flags target shorter duration noise transients and are typically tailored to specific analyses. In O4b and O4c, their use continues to diverge between search classes: CBC searches largely avoid using CAT2 vetoes, relying instead on

probabilistic DQ metrics, while short-duration unmodelled burst searches still employ CAT2 flags to veto periods of substantial contamination of astrophysical data. Because the latter class of searches are looking for generic excesses in strain coincident between detectors, their *DQ* demands are more restrictive compared to CBC searches.

5.2. Data quality for transient searches

Transient searches target signals with durations ranging from sub-second to minutes, including CBCs and unmodelled burst signals. These searches must balance sensitivity with robustness to transient noise artifacts.

5.2.1. Data quality for compact binary coalescence searches CBC searches in O4b and O4c continued the O4a strategy of relying on statistical, auxiliary-channel-based data quality information rather than traditional CAT2 segment vetoes [9]. The main DQ product generated for these searches was the iDQ timeseries [123], computed independently for each LIGO interferometer from auxiliary channel activity using the Ordered Veto List (OVL) algorithm [157]. For each strain sample, iDQ quantifies the evidence that a transient noise artifact is present, expressed as a false-alarm probability (FAP): the probability that a detector free of transient noise would produce auxiliary-channel activity at least as significant. A small FAP therefore corresponds to a confident identification of transient noise. The configuration and calibration of iDQ were unchanged from O4a [9], as was its consumption by the searches: although the low-latency iDQ output was distributed to all of the online CBC pipelines (PyCBC LIVE, GSTLAL, MBTA, and SPIIR), the PyCBC-based analyses were the only ones to ingest it, with the remaining pipelines relying on their own internal noise mitigation.

iDQ information was used in two forms, matching the two modes in which the PyCBC search operates. In low latency, PyCBC LIVE retained the O4a criterion of discarding candidates whose coalescence times fell within ± 1 s of any sample with $\text{FAP}(t) < 10^{-4}$, i.e., periods when transient noise was confidently identified [158, 159]. Offline, where longer stretches of data are available for ranking and calibrating vetoes, times satisfying $\log \mathcal{L}(t) \geq 5$, where $\mathcal{L}(t)$ is the iDQ likelihood ratio for the presence of transient noise, were expanded by ± 0.25 s to form a glitch flag. Rather than applying this flag as a hard veto, the offline PyCBC search folds it into its ranking statistic [124]: templates are grouped into bins by duration, and within each bin, candidates occurring during flagged time are down-ranked according to how strongly that bin’s single-detector triggers concentrate in flagged time relative to the bin’s mean trigger rate.

Figure 21 shows these relative trigger rates at LHO for a representative two-week interval of O4c, during which the glitch flag was active for 0.08% of the analysed livetime. For the shortest-duration templates, which are the most easily mimicked by transient noise, the trigger rate during flagged time exceeds the bin mean by more than two orders of magnitude: nearly 14% of that bin’s triggers occur within only 0.08% of the livetime. Candidates from these templates are therefore strongly down-ranked when they occur

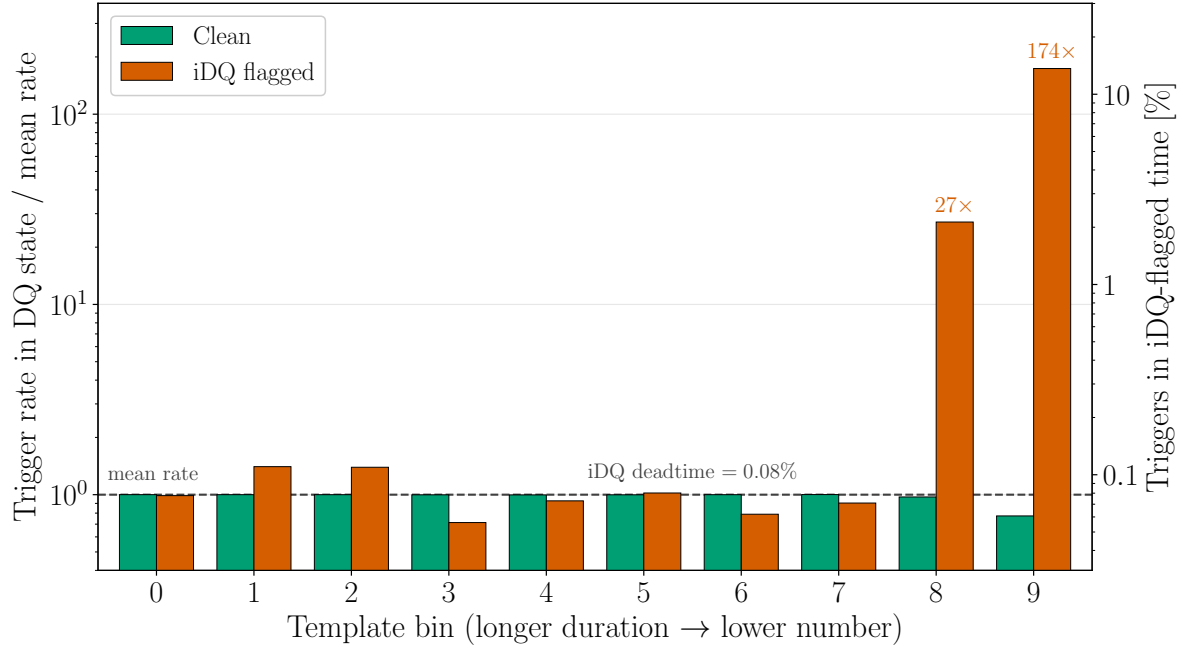


Figure 21: Impact of the offline iDQ glitch flag on the PyCBC search at LHO. For each PyCBC template-duration bin, bars give the rate of single-detector triggers recorded while the DQ is clean (green) and while the offline iDQ glitch flag is active (orange), normalised by the mean trigger rate in that bin. The dashed line marks the rate expected if triggers were distributed uniformly in time. Clean data sit on this line in every bin, whereas triggers during flagged time concentrate strongly in the shortest-duration templates, which are the most easily mimicked by transient noise. Candidates from these templates are consequently down-ranked when they occur during flagged time; candidates occurring in clean data are essentially unaffected. The figure is based on a representative two-week interval of O4c LHO data from June 24, 2025 to July 8, 2025.

during flagged time, while candidates in clean data—which trigger at the mean rate in every bin—are essentially unaffected, as are candidates from longer-duration templates, for which glitches are a weaker contaminant. In this way iDQ concentrates its down-weighting where instrumental glitches are most likely to be confused with genuine CBC signals.

No new auxiliary-channel-based CAT2 flag classes were introduced for CBC searches during O4b and O4c; the focus instead remained on monitoring iDQ performance across the run and on the event-level validation procedures described in Section 4.

5.2.2. Data quality for unmodelled transient searches Burst searches are broadly divided into two categories: short-duration and long-duration analyses. Short-duration searches target signals with durations of $\lesssim \mathcal{O}(1\text{s})$ and are especially susceptible to short glitches in the detector. Many, but not all, of these searches continue to utilize both CAT1 and CAT2 data-quality flags. Long-duration searches, which probe signals

lasting from several seconds to hundreds of seconds, are performed using both cWB and PySTAMPAS [160,161]. Being unmodelled, these searches make minimal assumptions on signal morphology and instead identify excess power coherent across the detector network in time-frequency maps of the strain data. Each pipeline is additionally run in several configurations that differ in their time-frequency decomposition, ranking statistic, or time-frequency pixel clustering algorithm, so as to remain sensitive to this broad range of possible signal morphologies. Because these signals persist over longer timescales, long-duration analyses are correspondingly more sensitive to extended spectral features such as persistent narrowband lines.

CAT2 flags for burst searches are constructed from auxiliary channels showing strong correlations with glitches in the strain data. In O4b and O4c, these include flags targeting very loud glitches (e.g., $\text{SNR} > 100$) as well as site-specific disturbances such as 60 Hz-related glitches observed at LLO [162]. The use of CAT2 flags in burst searches reduces the background rate of triggers while maintaining low deadtime (typically $\sim 0.1\%$).

Long-duration searches, particularly those targeting narrowband or quasi-monochromatic waveforms such as magnetar emission or gamma-ray burst plateau signals, are additionally sensitive to periods of narrowband line artifacts and transient line ring-ups. These features can mimic or obscure astrophysical signals in time-frequency analyses. In practice, analyses targeting these sources apply their own DQ measures, including removing the known narrowbands artifacts from the time-frequency space (PySTAMPAS) or applying decision tree classifiers to filter out triggers associated with line artifacts. The list of lines removed from the analysis is informed by the instrumental lines list compiled by DetChar investigations [13].

5.3. Data quality for persistent searches

Persistent gravitational-wave searches target signals that remain in band over long durations, including CWs and the SGWB. These analyses integrate data over long timescales and are therefore sensitive to non-stationarity and narrowband spectral artifacts that can accumulate over time.

Data quality efforts for persistent searches focus on mitigating transient glitches, slow variations in detector noise, and persistent or wandering spectral features of instrumental origin. While the specific procedures differ between CW and stochastic analyses, both rely on careful mitigation of time-domain frequency-domain contamination to ensure unbiased results.

The following sections summarize the data quality treatments applied in CW and stochastic searches during O4b and O4c. For a comprehensive discussion of CW data quality products and methodology during O4, see [13].

5.3.1. Data quality for continuous wave searches

Self-gated strain Glitches introduce noise into the detectors that degrade the sensitivity of CW searches. For this reason, a procedure known as “self-gating” is applied to the strain data prior to ingestion by CW analysis pipelines in which loud glitches are identified and zeroed out in the time domain. The removal of loud glitches via gating reduces the noise floor primarily below 500 Hz, thereby improving sensitivity to CW signals. The self-gating algorithm is based on the same iterative threshold selection procedure used in O4a, the implementation of which is described in detail in [163].

Lines lists Catalogs of line and comb artifacts (“lines lists”) observed in run-averaged spectra were produced for each detector [13]. These lists are an important DQ product used by CW searches, primarily for the purposes of vetoing non-astrophysical signal candidates. The lines lists are assembled via inspection of run-averaged high-resolution spectra, where the spectra are generated from Hann-windowed and self-gated 7200 s $h(t)$ segments. This results in a detailed spectrum with ~ 0.13889 mHz frequency resolution suitable for identifying persistent spectral contamination. Each line and comb artifact is identified and recorded through a combination of automated line-tagging tools and manual visual inspection of the run-averaged spectra.

For each detector, we recorded artifacts in either a “vetted” list or an “unvetted list”. The vetted list includes artifacts determined to be almost certainly non-astrophysical. This determination is usually based on correlating the appearance or disappearance of the artifact with detector hardware changes; an exception to this are combs, which are automatically added to the vetted list due to being inconsistent with CW emission models.

By the end of the O4, the total number of vetted line artifacts (counting individual comb teeth) in LHO was 2399, accounting for $\sim 17.21\%$ of the 10-2000 Hz observing band; in LLO, the line count was 449, accounting for $\sim 6.37\%$ of the observing band. Most of the vetoed band fraction for LHO is due to the excessive line contamination flanking the violin modes; without counting these contaminated regions, the vetoed band percentage at LHO is comparable to that of LLO.

5.3.2. Data quality for stochastic searches

Time-domain DQ measures: gating and non-stationarity cuts. Time-domain cleaning for stochastic searches begins with auto-gating within the PYGWB workflow [164] to suppress short-duration transients. Outliers in the whitened strain are identified using an amplitude threshold, and an inverse Planck-taper window is applied to smoothly suppress the data around these excursions. This removes glitches while avoiding sharp spectral artifacts that could otherwise bias segment-level noise estimates and causing unnecessary data loss. In O4b and O4c, the fraction of gated data was $\lesssim 0.1\%$ for each interferometer.

After gating, the data are divided into segments, typically 192 s in duration. For

each segment, the stochastic estimator uncertainty σ_Y

$$\Delta\sigma_i = \frac{\sigma_{Y,i} - \text{median}(\sigma_Y)}{\text{median}(\sigma_Y)} \quad (1)$$

is computed. Segments with $|\Delta\sigma_i|$ above a chosen threshold are excluded. Because σ_Y depends on the assumed spectral shape of the SGWB, the non-stationarity cut is applied separately for $\alpha \in \{-5, 0, 3, 5\}$ [34, 164], so different segments may be excluded depending on the spectral-index search.

The fraction of segments removed by non-stationarity cuts is less than $\sim 10\%$ of the total coincident observing time between the 2 LIGO detectors. Without gating, the fraction of segments removed by non-stationarity cuts would have been $\sim 27\%$, demonstrating the robustness of the gating procedure in dealing with short-duration transients and preventing unnecessary data loss.

Frequency-domain DQ measures: coherence and notching. Frequency-domain DQ measures target spectral artifacts that can introduce spurious correlated power between detectors and bias the cross-correlation estimator. They are cataloged in a DQ product known as a notch list, which defines list of frequency bins to be excluded from the analysis. As SGWB searches combine the information from a broad frequency range, the notch list is optimized to remove contaminated bins that could bias the estimator while preserving as much bandwidth as possible.

The notch list combines three components. Known instrumental features – calibration lines, simulated CW signals, power-mains harmonics, and mechanical resonances – are identified by direct inspection of run-averaged spectra. Bins are also flagged as statistically significant coherence outliers, quantified relative to the expected distribution under the null hypothesis of uncorrelated noise. Finally, bins with excess temporal variability that would not necessarily stand out in a run-averaged coherence or Power Spectral Density (PSD) spectrum are added, as described below. Figure 22 demonstrates the coherence-based component using the timeshifted data of LHO and LLO.

In the coherence spectra, most frequency bins lie near the statistical floor, while a small number exceed the threshold and appear as outliers. They include a simulated CW signal at 52.81 Hz, excess power near the 60 Hz power mains, a tooth of the 100Hz comb, one of the harmonics of the beamsplitter violin mode, as well as the cluster of the DuoTone lines at 960 Hz. After applying notch lists, the high-coherence tail is significantly suppressed, demonstrating the removal of correlated spectral artifacts.

Additionally, a complementary approach is to identify frequency bins exhibiting excess temporal variability in their PSDs. Such variability can indicate non-stationary or intermittently active noise sources, which may not be fully captured in run average spectra. This information is used to construct “strict” notches, where frequency bins exhibiting excess temporal variability in the detector PSDs,

$$\text{StdRatio}(f) = \text{std} \left[\frac{\text{PSD}(f)}{\text{median}_t \text{PSD}(f)} \right], \quad (2)$$

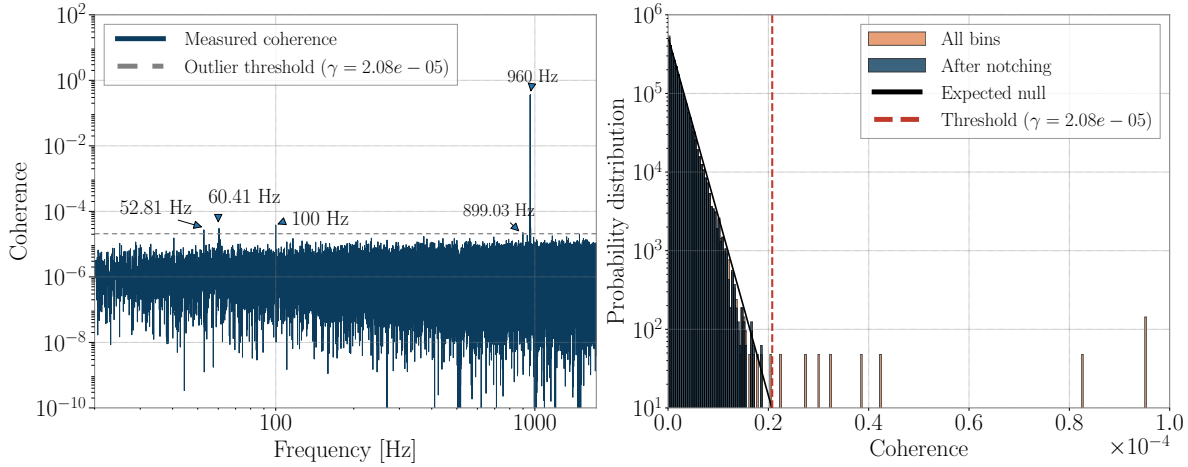


Figure 22: Left: coherence spectrum between LHO and LLO strain data over the analysis band. The dashed line indicates the statistical threshold corresponding to a fixed FAP under the null hypothesis of uncorrelated noise. Prominent outliers correspond to known instrumental features. Right: distribution of coherence values across all frequency bins. The gray histogram shows all bins, while the shaded histogram shows the distribution after applying notch lists. The solid curve indicates the expected null distribution. Notching significantly suppresses the high-coherence tail.

are flagged as non-stationary. A smooth local baseline for $\text{StdRatio}(f)$ is estimated by identifying peak-like structures that rise above a running median-filtered version of the spectrum, masking those regions, and interpolating linearly across the remaining bins, so that the baseline is representative of the typical variability within a local frequency band rather than a single global threshold. Frequency bins whose $\text{StdRatio}(f)$ exceeds this baseline by a factor of 1.5 are flagged as strict-notch candidates. Each candidate band is then manually vetted: bands that appear in only one detector and coincide with instrumental origins are retained, while marginal candidates with $\text{StdRatio}(f)$ close to threshold and no corresponding structure in the run-averaged PSD are excluded. Figure 23 shows the resulting variability spectra for the LIGO detectors. Most frequencies lie near a baseline consistent with stationary noise, while peaks correspond to non-stationary spectral features. Low-frequency scattered-light glitches at LLO, discussed in section 3.3.1, elevate its variability spectrum below ~ 40 Hz, while the wandering ~ 550 – 640 Hz narrowband feature described in Section 3.3.3 dominates LLO’s excess variability at higher frequencies and was among the strict-notch bands designed to prevent a recurrence of the spurious excess in the power-law posterior at large spectral index during the O4a search for SGWB [34, 165]. This method is particularly effective at identifying bands affected by wandering or intermittent artifacts that may not appear as strong coherence outliers but can still bias the estimator. In total, strict notching contributes an additional 4.5% of the $[20, 1726]$ Hz analysis band beyond standard notching procedures, bringing the total fraction of the band removed to 22.8% for O4b and O4c data up to the end of March 2025.

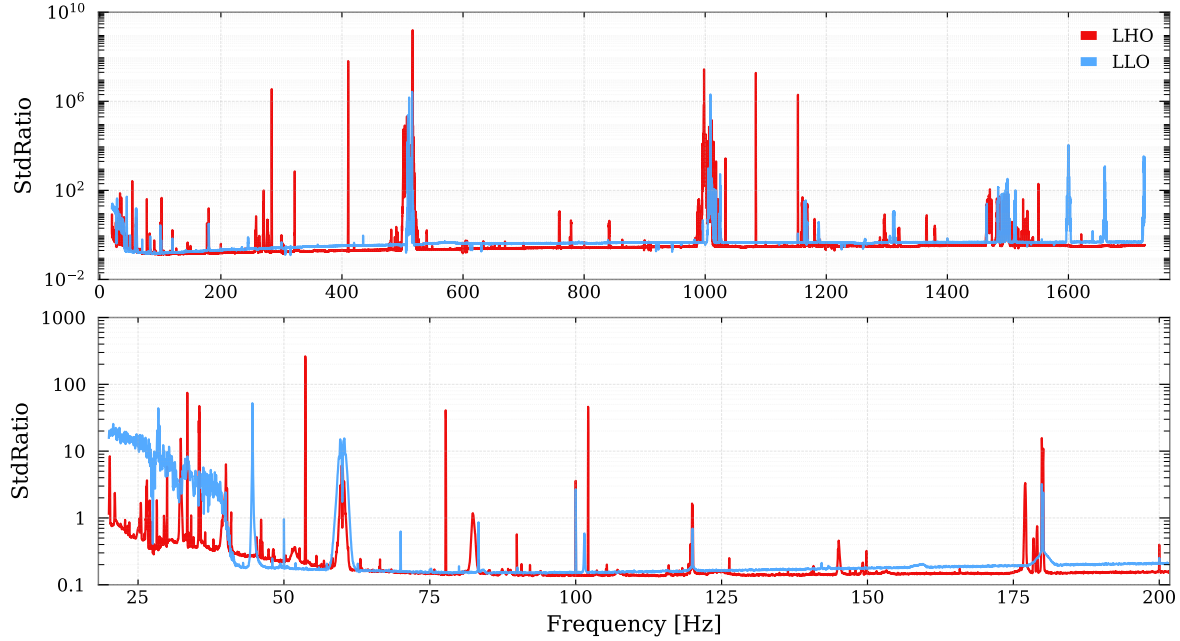


Figure 23: Standard deviation of normalized PSDs for LHO (red) and LLO over O4b and O4c up to the mid-O4c commissioning break when LLO’s GV was replaced.

6. Future Outlook

During O4, the LIGO DetChar group monitored data from the two LIGO detectors, characterized noise artifacts, and provided feedback to detector scientists on mitigation efforts. Data quality products were produced and employed by GW searches and source parameter estimation in order to improve search sensitivity, detection efficiency, and precision of astrophysical source parameter estimates. O4 was the longest and most sensitive run to date, and the LIGO Laboratory has, unlike during each of O1–O3, operated the two detectors through several epochs of observing and commissioning breaks during which the detectors have evolved. During the commissioning breaks following O4a, and occasionally during required breaks in observing during O4b and O4c to repair damaged components, the LIGO detectors adopted improved hardware and configuration changes in order to improve their overall performance.

Long observing runs offer unique opportunities to measure and investigate long-term changes of detector noise that are caused by environmental factors or detector configuration changes. Overall, we have observed improvements in some areas but some degradations in others. While detections of GW signals from CBC sources continues apace, care must be taken with each signal, as evidenced by the Data Quality Report effort made for each detection and parameter estimation follow-up. Other classes of GW signals have not yet been detected ranging from unmodeled, short duration “burst” sources to persistent sources from spinning neutron stars and SGWB signals. Identifying noise sources and, where possible, mitigating the sources or coupling mechanisms are crucial for detecting these new classes of signals.

We have described the changes made to the LIGO instruments and studies carried out during O4b and O4c to investigate various noise sources impacting the LIGO detectors. Some of these studies were initiated because of changes in detector performance while others were initiated because of detector configuration changes. For low-latency, online transient GW searches, we have summarized the Data Quality Report, a standard approach for quality validation that is carried out rapidly following the appearance of a candidate signal discovered in low-latency analyses. This procedure has proven successful to streamline follow-up investigations and rapidly determine if GW candidate signals are impacted by noise artifacts. Modest upgrades to the Data Quality Report since the initial deployment in O4a have further improved the utility of this framework.

Low-latency, online transient GW searches, follow-up GW source parameter estimates, and high-latency, offline GW searches rely on data quality products produced as part of the DetChar group efforts. These products are essential in order to avoid analyzing data of poor quality that could impact search results. The Data Quality Report is an example of a data quality product developed for O4 as part of the broader DetChar group activities which automates low-latency detector data quality assessment triggered by transient GW candidates.

Further efforts to streamline other DetChar investigation, data quality product, and GW candidate follow-up procedures will be essential as detector sensitivity continues to improve, observing run durations increase in length, and new classes of GW signals are detected. While automation efforts will be essential in the years ahead, we expect that future detector improvements will reveal new noise sources and coupling pathways that may require development of new investigative tools and mitigation strategies. Integrating new tools with existing ones, maintaining these computational workflows, and being responsive to new developments requires an agile and dedicated team of personnel involved in LIGO DetChar activities.

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References

- [1] Abbott B P *et al* 2016 GW150914: First results from the search for binary black hole coalescence with Advanced LIGO *Phys. Rev. D* **93** 122003 (*Preprint* arXiv:1602.03839)
- [2] Abbott B P *et al* (LIGO-Virgo Collaboration) 2019 GWTC-1: A Gravitational-Wave Transient Catalog of Compact Binary Mergers Observed by LIGO and Virgo during the First and Second Observing Runs *Phys. Rev. X* **9** 031040 (*Preprint* arXiv:1811.12907)
- [3] Abbott R *et al* (LIGO-Virgo Collaboration) 2021 GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo during the First Half of the Third Observing Run *Phys. Rev. X* **11** 021053 (*Preprint* arXiv:2010.14527)
- [4] Abbott R *et al* 2023 GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo during the Second Part of the Third Observing Run *Phys. Rev. X* **13** 041039 (*Preprint* arXiv:2111.03606)
- [5] Abac A *et al* (LIGO-Virgo-KAGRA Collaboration) 2025 GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run (*Preprint* arXiv:2508.18082)
- [6] Abac A *et al* (LIGO-Virgo-KAGRA Collaboration) 2026 GWTC-5.0: Observations from the Second Part of the Fourth LIGO-Virgo-KAGRA Observing Run and Updates to the Gravitational-Wave Transient Catalog (*Preprint* arXiv:2605.27225)
- [7] Acernese F *et al* 2015 Advanced Virgo: a second-generation interferometric gravitational wave detector *Classical Quantum Gravity* **32** 024001 (*Preprint* arXiv:1408.3978)
- [8] Kagra Collaboration 2019 KAGRA: 2.5 generation interferometric gravitational wave detector *Nat. Astron.* **3** 35–40 (*Preprint* arXiv:1811.08079)
- [9] Soni S *et al* 2025 LIGO Detector Characterization in the first half of the fourth Observing run *Classical Quantum Gravity* **42** 085016 (*Preprint* arXiv:2409.02831)
- [10] Acernese F *et al* 2023 Virgo detector characterization and data quality: results from the O3 run *Classical Quantum Gravity* **40** 185006 (*Preprint* arXiv:2210.15633)
- [11] Akutsu T *et al* 2021 Overview of KAGRA: Calibration, detector characterization, physical environmental monitors, and the geophysics interferometer *Progress of Theoretical and Experimental Physics* **2021** 05A102 (*Preprint* arXiv:2009.09305)
- [12] Davis D and Capote E 2025 Guiding interferometer improvements with the frequency-dependent inspiral range *Phys. Rev. D* **111** 122005 (*Preprint* arXiv:2502.07253)
- [13] Goetz E *et al* 2026 Narrow spectral artifact investigation and mitigation in LIGO data from the fourth LIGO-Virgo-KAGRA observing run (*Preprint* arXiv:2606.05959)
- [14] Davis D *et al* 2021 LIGO detector characterization in the second and third observing runs *Classical Quantum Gravity* **38** 135014 (*Preprint* arXiv:2101.11673)

- [15] Nuttall L K 2018 Characterizing transient noise in the LIGO detectors *Philos. Trans. R. Soc. London, Ser. A* **376** 20170286 (*Preprint* arXiv:1804.07592)
- [16] Abbott B P and other 2016 Characterization of transient noise in Advanced LIGO relevant to gravitational wave signal GW150914 *Classical Quantum Gravity* **33** 134001 (*Preprint* arXiv:1602.03844)
- [17] Zevin M *et al* 2017 Gravity Spy: integrating advanced LIGO detector characterization, machine learning, and citizen science *Classical Quantum Gravity* **34** 064003 (*Preprint* arXiv:1611.04596)
- [18] Glanzer J *et al* 2023 Data quality up to the third observing run of advanced LIGO: Gravity Spy glitch classifications *Classical Quantum Gravity* **40** 065004 (*Preprint* arXiv:2208.12849)
- [19] Payne E, Hourihane S, Golomb J, Udall R, Davis D and Chatziioannou K 2022 Curious case of GW200129: Interplay between spin-precession inference and data-quality issues *Phys. Rev. D* **106** 104017 (*Preprint* arXiv:2206.11932)
- [20] Lecoecue Y, McIver J, Knee A M, Udall R, Rink K, Hourihane S, Miller S J, Chatziioannou K, Massinger T and Davis D 2026 Coalescing Compact Binary Parameter Estimation with Gravitational Waves in the Presence of non-Gaussian Transient Noise (*Preprint* arXiv:2604.07668)
- [21] Udall R, Bini S, Chatziioannou K, Davis D, Hourihane S, Lecoecue Y, McIver J and Miller S 2026 Inferring the spins of merging black holes in the presence of data-quality issues *Phys. Rev. D* **113** 042005 (*Preprint* arXiv:2510.05029)
- [22] Karki S *et al* 2016 The Advanced LIGO photon calibrators *Rev. Sci. Instrum.* **87** 114503 (*Preprint* arXiv:1608.05055)
- [23] Essick R, Mo G and Katsavounidis E 2021 A coincidence null test for Poisson-distributed events *Phys. Rev. D* **103** 042003 (*Preprint* arXiv:2011.13787)
- [24] Pem central URL <https://pem.ligo.org/>
- [25] Ganapathy D *et al* 2023 Broadband Quantum Enhancement of the LIGO Detectors with Frequency-Dependent Squeezing *Phys. Rev. X* **13** 041021
- [26] Schwartz E *et al* 2020 Improving the robustness of the advanced LIGO detectors to earthquakes *Classical Quantum Gravity* **37** 235007 (*Preprint* arXiv:2007.12847)
- [27] Capote E and Warner J Trying High Bandwidth control for earthquake <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=85740>
- [28] Pham H Start testing EQ control with ASC cut-off <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=78473>
- [29] Capote E *et al* 2025 Advanced LIGO detector performance in the fourth observing run *Phys. Rev. D* **111** 062002 (*Preprint* arXiv:2411.14607)
- [30] Matichard F *et al* 2015 Seismic isolation of Advanced LIGO: Review of strategy, instrumentation and performance *Classical Quantum Gravity* **32** 185003 (*Preprint* arXiv:1502.06300)
- [31] Nguyen P, Schofield R M S, Effler A *et al* 2021 Environmental noise in advanced LIGO detectors *Classical Quantum Gravity* **38** 145001 (*Preprint* arXiv:2101.09935)
- [32] Janssens K, Lawrence J, Effler A, Schofield R M S, Lalleman M, Betzwieser J, Christensen N, Coughlin M W, Driggers J C, Helmling-Cornell A F *et al* 2025 Coherent injection of magnetic noise and its impact on gravitational-wave searches *Phys. Rev. D* **112** 102003 (*Preprint* arXiv:2505.11903)
- [33] Venikoudis S, De Lillo F, Janssens K, Suresh J and Bruno G 2025 Impact of correlated magnetic noise on directional stochastic gravitational-wave background searches *Phys. Rev. D* **111** 082005 (*Preprint* arXiv:2411.11746)
- [34] Abac A G *et al* (LIGO-Virgo-KAGRA Collaboration) 2025 Upper Limits on the Isotropic Gravitational-Wave Background from the first part of LIGO, Virgo, and KAGRA's fourth Observing Run (*Preprint* arXiv:2508.20721)
- [35] Pham K, Schofield R and Campos C May 15 LEMI calibration <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=84924>
- [36] Effler A, Pham K, Guidry T, MacDonald T and Connolly G Vault Tests for LEMI magnetometers

- calibration <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=79137>
- [37] Bossilkov V Greater angular offload to L1 <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=78903>
 - [38] Bossilkov V Summary of Commissioning Changes <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=78942>
 - [39] Driggers J *et al* IFO very unhappy, something likely wrong with alignment, AS_C needs strange offsets <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=77368>
 - [40] Dwyer S *et al* we suspect that something in OFI has changed <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=77392>
 - [41] Dwyer S Summary of alogs related to changes in our output arm this week <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=77427>
 - [42] Sanchez A, Kijbunchoo N and Xu V Re-aligning SQZ to current AS alignment 4.5dB, required big ZM moves <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=77400>
 - [43] Kawabe K, Dwyer S, Crouch R and Abouelfettouh I DARM offset causing locklosses again <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=79082>
 - [44] Dwyer S some symptoms that seem similar to April 22nd/23rd output change <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=79101>
 - [45] Kawabe K and Dwyer S KTP wedge chipped, back up spare installed and ready to go to chamber <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=79326>
 - [46] Dwyer S, Kawabe K, Driggers J and Compton C FS Assmembly Reinstalled in OFI <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=79363>
 - [47] Gray C, Patane O, Short R and Dwyer S H1 Back To Observing!!! <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=79690>
 - [48] Oberling J, Short R, Dwyer S, Sanchez A, Xu V, Capote C and Compton C Overview of PSL Story so far <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=81193>
 - [49] Driggers J Comment to: Test setup of spare NPRO does not seem to see glitches; beginning process to swap to spare NPRO <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=80837>
 - [50] Short R and Oberling J PSL NPRO Swap: It Lives! (WP 12155) <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=80908>
 - [51] Compton C Comparison of PSL glitches before locklosses tagged: glitches dramatically changed after NPRO swap <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=81386>
 - [52] Xu V and Oberling J PSL NPRO Crystal Temperature Tuning <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=81107>
 - [53] Oberling J and Short R PSL NPRO Swap: Here We Go Again (WP 12210) <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=81409>
 - [54] Robinet F, Arnaud N, Leroy N, Lundgren A, Macleod D and McIver J 2020 Omicron: A tool to characterize transient noise in gravitational-wave detectors *SoftwareX* **12** 100620 (*Preprint* arXiv:2007.11374)
 - [55] Brown J C 1991 Calculation of a constant Q spectral transform *J. Acoust. Soc. Am.* **89** 425–434
 - [56] Davis D, Massinger T, Lundgren A, Driggers J C, Urban A L and Nuttall L 2019 Improving the sensitivity of Advanced LIGO using noise subtraction *Classical Quantum Gravity* **36** 055011 (*Preprint* arXiv:1809.05348)
 - [57] Dartez L, Dwyer S and Vajente, G New DARM offloading reduces non-stationarity by reducing ESD drive RMS. <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=74939>
 - [58] Cabero M *et al* 2019 Blip glitches in Advanced LIGO data *Classical Quantum Gravity* **36** 155010 (*Preprint* arXiv:1901.05093)
 - [59] Nandi D, Effler A, Soni S, Aira Ferreira T, Schofield R, Pham H, O’Hanlon T, Frolov V V and González G 2026 Scattered light noise at LIGO Livingston Observatory during O4 (*Preprint* arXiv:2605.14143)
 - [60] Ferreira T and Gonzalez G Significant discrepancy between the spectrograms from

- GDS_CALIB_STRAIN and GDS_CALIB_STRAIN_CLEAN/NOLINES <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=69626>
- [61] Ferreira T and Gonzalez G Omicron at GDS_CALIB_STRAIN and GDS_CALIB_STRAIN_NOLINES <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=69850>
- [62] Soni S *et al* 2021 Reducing scattered light in LIGO's third observing run *Classical Quantum Gravity* **38** 025016 (*Preprint* arXiv:2007.14876)
- [63] Schofield R CER ACs and main HVAC likely couple to DARM in input arm, possibly through MCA2 baffle <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=74175>
- [64] Schofield R Confirmation from laser vibrometry and beating shakers: the MC baffle by HAM3 has been producing noise in DARM and the one by HAM2 produces noise when 18 Hz vibration is increased by about 5; also, suggestions for mitigation during the January break <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=74772>
- [65] Schofield R Evaluation of stray light work done during the break: desired reductions in vibration coupling attained for EX cryobaffle, and for input arm - after adjustment of ITMY CP yaw <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=76969>
- [66] Connolly G Hunting 30-40 Hz peaks with portable accelerometer <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=79546>
- [67] Callos S and Schofield R 24-30 Hz Range Noise in DARM Identification <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=80182>
- [68] Connolly G Identification of 30-40 Hz peaks in DARM <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=80655>
- [69] Schofield R, McCarthy R, Guidry T, Connolly G, Mannix B and Olson J 35 Hz peak in DARM greatly reduced by changing the frequency of the OSB HVAC drive. <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=82986>
- [70] Callos S and Schofield R 20-40 Hz HVAC Noise <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=86257>
- [71] Mérou J, Calafat A, Effler A, Dwyer S, Schofield R and Driggers J Witness channels of the set of near-30Hz and near-100Hz combs at LHO <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=87414>
- [72] Aasi J *et al* (LIGO Scientific Collaboration) 2015 Advanced LIGO *Classical Quantum Gravity* **32** 074001 (*Preprint* arXiv:1411.4547)
- [73] Lockerbie N A, Carbone L, Shapiro B, Tokmakov K V, Bell A and Strain K A 2011 First results from the 'Violin-Mode' tests on an advanced LIGO suspension at MIT *Classical Quantum Gravity* **28** 245001
- [74] Starkman T Effect of violin mode amplitude on narrow lines near the violin modes <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=71501>
- [75] Starkman T Asymmetry in narrow line contamination in bands around violin mode frequencies <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=71800>
- [76] Goetz E *et al* O3 lines and combs in found in self-gated C01 data <https://dcc.ligo.org/LIGO-T2100200/public>
- [77] Goetz E *et al* O4a lines and combs in found in self-gated C00 cleaned data <https://dcc.ligo.org/LIGO-T2400204/public>
- [78] Neunzert A CAL_AWG_LINES extra calibration lines create many narrow artifacts below 100 Hz <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=71964>
- [79] Neunzert A, Goetz E, Knee A, Collier T and Marceau A Deeper study of a 2023 test shows calibration lines + violin modes = more line contamination than previously understood <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=79825>
- [80] Neunzert A Relationship between violin mode height and narrow spectral artifact contamination, revisited <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=82320>
- [81] Goetz E, Knee A and Neunzert A O4a lines statistics and most problematic artifacts <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=82320>

- [//alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=79579](https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=79579)
- [82] Calafat A, Mérou J, Dwyer S, Schofield R and Driggers J Intermodulation lines and β estimation in H1:OMC-DCPD.SUM_OUT_DQ <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=87923>
 - [83] Schofield R and Dwyer S Update on grounding noise studies <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=67075>
 - [84] Effler A PEM Injection Status <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=64609>
 - [85] Effler A Commissioning tests today <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=67149>
 - [86] Schofield R ETMX bias that minimizes grounding noise coupling hasnt changed much since January, and PEM injections at 60W <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=72118>
 - [87] Schofield R, Dwyer S and Compton C ETMX bias sweep gives significantly different minimum in electronics ground noise coupling after charge change: from 150 V to new value of 58 V (1.5 bias offset) <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=78194>
 - [88] Schofield R Slow changes of ITMY ESD bias during observation <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=78734>
 - [89] Schofield R Slow bias changes during observation completed; New ESD bias settings: ITMX 0V, ITMY -40V (-4.0 in offset window) <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=78925>
 - [90] Helmling-Cornell A F, Nguyen P, Schofield R M S and Frey R 2024 Automated evaluation of environmental coupling for Advanced LIGO gravitational wave detections *Classical Quantum Gravity* **41** 145003 (*Preprint* arXiv:2312.00735)
 - [91] Berger B K *et al* 2023 Searching for the causes of anomalous Advanced LIGO noise *Appl. Phys. Lett.* **122** 184101
 - [92] Effler A, Schofield R M S, Frolov V V, González G, Kawabe K, Smith J R, Birch J and McCarthy R 2015 Environmental influences on the LIGO gravitational wave detectors during the 6th science run *Classical Quantum Gravity* **32** 035017 (*Preprint* arXiv:1409.5160)
 - [93] Short R, Glanzer J, Helmling-Cornell A, Effler A and Schofield R PEM injections successfully completed - will inform HVAC experiments <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=69745>
 - [94] Cahillane C Comment to: Another failed attmpt to switch OMC ASC to dither alignment <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=64609>
 - [95] Callos S, Connolly G, Schofield R, Short R and Campos C PEM November 2025 Injection Coupling Analysis Complete <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=89929>
 - [96] Washimi T, Yokozawa T, Tanaka T, Itoh Y, Kume J and Yokoyama J 2021 Method for environmental noise estimation via injection tests for ground-based gravitational wave detectors *Classical Quantum Gravity* **38** 125005 (*Preprint* arXiv:2012.09294)
 - [97] Pham H, O'Hanlon T and Effler A BS HEPI injections imply large amount of scattered light in the vertex, from ITM elliptical baffles <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=78693>
 - [98] Nandi D and Ferreira T Comparing the rate of scattered light glitches pre and post vent <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=78226>
 - [99] Nandi D 2025 aLIGO LLO Logbook <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=75579>
 - [100] Pham H 2024 aLIGO LLO Logbook <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=72674>
 - [101] O'Hanlon T and Effler E EX Manifold Wrapped to try and affect 30 min oscillations <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=73481>

- [102] Frolov V and Effler E EX tapping tests and microseism drive <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=73288>
- [103] Berger B Wandering lines in PEM VMON channels at EX line up with BNS range oscillations <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=80232>
- [104] Berger B Time variation of wandering lines in PEM EX VMON ESDPOWER channels <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=80556>
- [105] Walker M, Agnew A F, Bidler J, Lundgren A, Macedo A, Macleod D, Massinger T J, Patane O and Smith J R 2018 Identifying correlations between LIGO's astronomical range and auxiliary sensors using lasso regression *Classical Quantum Gravity* **35** 225002 (*Preprint* arXiv:1807.02592)
- [106] Mullavey A Unlocking ALS end-station PLLs <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=75125>
- [107] Grant G and Glanzer J *Investigations of Binary Neutron Star Range Oscillations at LIGO Livingston* <https://dcc.ligo.org/LIGO-T2500220/public>
- [108] Kabagoz B and Effler A Low range, sqz on off test: It's not the squeezer <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=72583>
- [109] Glanzer J Comment to: Excess noise during range oscillation <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=72819>
- [110] Glanzer J Comment to: BRUCO comparison for range oscillations <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=78194>
- [111] Glanzer J Potential correlation between long duration range drops and CS temp <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=73009>
- [112] Brooks A Environmental coupling to the ITMs due to temperature excursions even with stabilized lab temperature <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=74834>
- [113] Iwaguchi S Comparison before and after changing AHU settings <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=78104>
- [114] Iwaguchi S AHU behavior and LVEA temperature after change of AHU1 setting <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=78563>
- [115] O'Hanlon T LVEA HVAC Controls Change <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=78617>
- [116] Kandhasamy S 2024 Wandering lines in h(t) around 580 Hz and other related lines <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=70869>
- [117] Macleod D, Goetz E, Ota I, Isi M, Massinger T, Davis D, Pitkin M, Altin P and Nitz A 2024 gwpy/gwsumm: 2.2.6 URL <https://doi.org/10.5281/zenodo.10999143>
- [118] Areeda J, Smith J, Lundgren A, Maros E, Macleod D and Zweizig J 2017 LigoDV-web: Providing easy, secure and universal access to a large distributed scientific data store for the LIGO scientific collaboration *Astron. Comput.* **18** 27–34 ISSN 2213-1337 URL <https://www.sciencedirect.com/science/article/pii/S2213133716301196>
- [119] Fisher R P, Hemming G, Bizouard M A, Brown D A, Couvares P F, Robinet F and Verkindt D 2021 DQSEGDB: A time-interval database for storing gravitational wave observatory metadata *SoftwareX* **14** 100677 (*Preprint* arXiv:2008.11316)
- [120] Vajente G 2023 BruCo: Brute-force coherence noise analysis URL <https://git.ligo.org/gabriele-vajente/bruco>
- [121] Smith J R, Abbott T, Hirose E, Leroy N, MacLeod D, McIver J, Saulson P and Shawhan P 2011 A hierarchical method for vetoing noise transients in gravitational-wave detectors *Classical Quantum Gravity* **28** 235005 (*Preprint* arXiv:1107.2948)
- [122] Isogai T, LIGO Scientific Collaboration and Virgo Collaboration 2010 Used percentage veto for LIGO and virgo binary inspiral searches *J. Phys. Conf. Ser. (J. Phys. Conf. Ser. vol 243)* (IOP) p 012005
- [123] Essick R, Godwin P, Hanna C, Blackburn L and Katsavounidis E 2020 iDQ: Statistical Inference of Non-Gaussian Noise with Auxiliary Degrees of Freedom in Gravitational-Wave Detectors

- (Preprint arXiv:2005.12761)
- [124] Davis D, Trevor M, Mozzon S and Nuttall L K 2022 Incorporating information from LIGO data quality streams into the PyCBC search for gravitational waves *Phys. Rev. D* **106** 102006 (Preprint arXiv:2204.03091)
 - [125] Talukder D 2017 LIGO Channel Activity Monitor URL <https://github.com/dipongkar/ligocam>
 - [126] Goetz E, Neunzert A and Suyamprakasam S 2025 Fscan URL <https://doi.org/10.5281/zenodo.18776397>
 - [127] Meyers P M 2018 *Cross-correlation searches for persistent gravitational waves with Advanced LIGO and noise studies for current and future ground-based gravitational-wave detectors* Ph.D. thesis Minnesota U.
 - [128] Hernandez G and E T 2023 *Stochmon: A LIGO Data Analysis Tool* Tech. Rep. T1400205 LSC URL <https://dcc.ligo.org/LIGO-T1400205/public>
 - [129] Usman S A, Nitz A H, Harry I W, Biwer C M, Brown D A, Cabero M, Capano C D, Canton T D, Dent T, Fairhurst S *et al* 2016 The pycbc search for gravitational waves from compact binary coalescence *Classical Quantum Gravity* **33** 215004
 - [130] Messick C, Blackburn K, Brady P, Brockill P, Cannon K, Cariou R, Caudill S, Chamberlin S J, Creighton J D, Everett R *et al* 2017 Analysis framework for the prompt discovery of compact binary mergers in gravitational-wave data *Phys. Rev. D* **95** 042001
 - [131] Aubin F, Brighenti F, Chierici R, Estevez D, Greco G, Guidi G M, Juste V, Marion F, Mours B, Nitoglia E *et al* 2021 The mbta pipeline for detecting compact binary coalescences in the third ligo–virgo observing run *Classical Quantum Gravity* **38** 095004
 - [132] Luan J, Hooper S, Wen L and Chen Y 2012 Towards low-latency real-time detection of gravitational waves from compact binary coalescences in the era of advanced detectors *Phys. Rev. D* **85** 102002 (Preprint arXiv:1108.3174)
 - [133] Hooper S, Chung S K, Luan J, Blair D, Chen Y and Wen L 2012 Summed parallel infinite impulse response filters for low-latency detection of chirping gravitational waves *Phys. Rev. D* **86** 024012 (Preprint arXiv:1108.3186)
 - [134] Marx E, Benoit W, Gunny A, Omer R, Chatterjee D, Venterea R C, Wills L, Saleem M, Moreno E, Raikman R *et al* 2025 Machine-learning pipeline for real-time detection of gravitational waves from compact binary coalescences *Phys. Rev. D* **111** 042010
 - [135] Skliris V, Norman M R and Sutton P J 2024 Toward real-time detection of unmodeled gravitational wave transients using convolutional neural networks *Phys. Rev. D* **110** 104034
 - [136] Mishra T, O’Brien B, Szczepańczyk M, Vedovato G, Bhaumik S, Gayathri V, Prodi G, Salemi F, Milotti E, Bartos I *et al* 2022 Search for binary black hole mergers in the third observing run of advanced ligo-virgo using coherent waveburst enhanced with machine learning *Phys. Rev. D* **105** 083018
 - [137] Cornish N J, Littenberg T B, Bécsy B, Chatziioannou K, Clark J A, Ghonge S and Millhouse M 2021 BayesWave analysis pipeline in the era of gravitational wave observations *Phys. Rev. D* **103** 044006 (Preprint arXiv:2011.09494)
 - [138] Di Renzo F, Fidecaro F, Razzano M and Sorrentino N 2024 BRiSTOL – a Band-limited RMS Stationarity Test Tool for Gravitational Wave Data (Preprint arXiv:2401.15392)
 - [139] Vazsonyi L and Davis D 2023 Identifying glitches near gravitational-wave signals from compact binary coalescences using the Q-transform *Classical Quantum Gravity* **40** 035008 (Preprint arXiv:2208.12338)
 - [140] Mozzon S, Nuttall L K, Lundgren A, Dent T, Kumar S and Nitz A H 2020 Dynamic normalization for compact binary coalescence searches in non-stationary noise *Classical Quantum Gravity* **37** 215014 (Preprint arXiv:2002.09407)
 - [141] Macas R and Lundgren A 2023 Sensitive test of non-Gaussianity in gravitational-wave detector data *Phys. Rev. D* **108** 063016 (Preprint arXiv:2306.09019)
 - [142] Álvarez-López S, Liyanage A, Ding J, Ng R and McIver J 2024 GSpyNetTree: a signal-vs-

- glitch classifier for gravitational-wave event candidates *Classical Quantum Gravity* **41** 085007 (*Preprint* arXiv:2304.09977)
- [143] Álvarez-López S, Chan M L, Herbst F S, Raghunathan D, Ahuja A, Liyanage A, Ding J, Garcia-Varela A, Ng R and McIver J 2026 GSPyNetTree-O4: an event validation tool used in the fourth LIGO-Virgo-KAGRA observing run (*Preprint* arXiv:2607.25208)
 - [144] Davis D *et al* 2026 Rapid data quality investigations of gravitational-wave events with the Data Quality Report Builder toolkit *arXiv e-prints* arXiv:2605.16183 (*Preprint* arXiv:2605.16183)
 - [145] Acernese F, Agathos M, Ain A, Albanesi S, Allocca A, Amato A, Andrade T, Andres N, Andrés-Carcasona M, Andrić T *et al* 2023 Virgo detector characterization and data quality: tools *Classical Quantum Gravity* **40** 185005
 - [146] Hourihane S and Chatziioannou K 2025 Glitches far from transient gravitational-wave events do not bias inference *Phys. Rev. D* **112** 084006 (*Preprint* arXiv:2506.21869)
 - [147] Cornish N J and Littenberg T B 2015 Bayeswave: Bayesian inference for gravitational wave bursts and instrument glitches *Classical Quantum Gravity* **32** 135012 (*Preprint* arXiv:1410.3835)
 - [148] Ashton, G *et al* 2024 CBCFlow <https://cbc.docs.ligo.org/projects/cbcflow/index.html>
 - [149] Ligo Scientific Collaboration, VIRGO Collaboration and Kagra Collaboration 2025 LIGO/Virgo/KAGRA S250108ha: Retraction of GW compact binary merger candidate *GRB Coordinates Network* **38856** 1
 - [150] Ligo Scientific Collaboration, VIRGO Collaboration and Kagra Collaboration 2024 LIGO/Virgo/KAGRA S240420aw: Retraction of GW compact binary merger candidate *GRB Coordinates Network* **36190** 1
 - [151] Ligo Scientific Collaboration, VIRGO Collaboration and Kagra Collaboration 2024 LIGO/Virgo/KAGRA S241104a: Retraction of GW compact binary merger candidate *GRB Coordinates Network* **38070** 1
 - [152] Ligo Scientific Collaboration, VIRGO Collaboration and Kagra Collaboration 2024 LIGO/Virgo/KAGRA S240624cd: Retraction of GW unmodeled transient candidate *GRB Coordinates Network* **36747** 1
 - [153] Helmling-Cornell A 2025 *Noise Mitigation in Searches for Gravitational Waves from Compact Binaries and Cosmic Strings* Ph.D. thesis University of Oregon
 - [154] Arnaud N 2023 LIGO and Virgo detector characterization and data quality: Contributions to the O3 run and preparation for O4 *Nucl. Instrum. Methods Phys. Res. A* **1048** 167945
 - [155] Abac A *et al* 2026 Open Data from LIGO, Virgo, and KAGRA through the Second Part of the Fourth Observing Run (*Preprint* arXiv:2605.27090)
 - [156] Evans M *et al* 2015 Observation of Parametric Instability in Advanced LIGO *Phys. Rev. Lett.* **114** 161102 (*Preprint* arXiv:1502.06058)
 - [157] Essick R, Blackburn L and Katsavounidis E 2013 Optimizing vetoes for gravitational-wave transient searches *Classical Quantum Gravity* **30** 155010 (*Preprint* arXiv:1303.7159)
 - [158] Dal Canton T, Nitz A H, Gadre B, Cabourn Davies G S, Villa-Ortega V, Dent T, Harry I and Xiao L 2021 Real-time Search for Compact Binary Mergers in Advanced LIGO and Virgo's Third Observing Run Using PyCBC Live *Astrophys. J.* **923** 254 (*Preprint* arXiv:2008.07494)
 - [159] Trevor M, Cabourn Davies G S, Dal Canton T, Dent T, Harry I, Hoang S and Tolley A 2026 PyCBC Live Search for Compact Binary Mergers in Advanced LIGO and Virgo's Fourth Observing Run (*Preprint* arXiv:2606.07679)
 - [160] Drago M *et al* 2021 coherent WaveBurst, a pipeline for unmodeled gravitational-wave data analysis *SoftwareX* **14** 100678 (*Preprint* arXiv:2006.12604)
 - [161] Macquet A, Bizouard M A, Christensen N and Coughlin M 2021 Long-duration transient gravitational-wave search pipeline *Phys. Rev. D* **104** 102005 (*Preprint* arXiv:2108.10588)
 - [162] Lundgren A Electrical glitches remain at EY after ESD transitioned to EX <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=64377>
 - [163] Davis D, Neunzert A, Goetz E, Riles K, Wette K and Lalleman M Self-gating of O4a h(t) for use in continuous-wave searches <https://dcc.ligo.org/LIGO-T2400003/public>

- [164] Renzini A I *et al* 2023 pygwb: A Python-based Library for Gravitational-wave Background Searches *Astrophys. J.* **952** 25 (*Preprint* arXiv:2303.15696)
- [165] Abac A *et al* In preparation